Component : xcand_top_comp

Description : **MemoryMap ExportedMemMap :** MemoryMap describing the whole flattened system.

Block : xcand_mh_inst_0__AXI_REG__0__xcand_mh_creg

Address	Name	Description	Reset
0x000	VERSION	Release Identification Register This register is protected by a register bank CRC defined in CRC_REG register.	0x0560 0000
0x004	MH_CTRL	Message Handler Control register	0x0000 0000
0x008	MH_CFG	Message Handler Configuration register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0700
0x00C	MH_STS	Message Handler Status register	0x0000 0000
0x010	MH_SFTY_CFG	Message Handler Safety Configuration register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x014	MH_SFTY_CTRL	Message Handler Safety Control register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000

0x018	RX_FILTER_MEM_ADD	RX Filter Base Address register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x01C	TX_DESC_MEM_ADD	TX Descriptor Base Address register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x020	AXI_ADD_EXT	AXI address extension register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x024	AXI_PARAMS	AXI parameter register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x028	MH_LOCK	Message Handler Lock register	0x0000 0000
0x100	TX_DESC_ADD_PT	TX descriptor current address pointer register	0x0000 0000
0x104	TX_STATISTICS	TX Message Counter register	0x0000 0000
0x108	TX_FQ_STS0	TX FIFO Queue Status register	0x0000 0000
0x10C	TX_FQ_STS1	TX FIFO Queue Status register	0x0000 0000
0x110	TX_FQ_CTRL0	TX FIFO Queue Control register 0	0x0000 0000
0x114	TX_FQ_CTRL1	TX FIFO Queue Control register 1 This register is only accessible in write mode if the unlock key sequence has been performed prior to write	0x0000 0000
0x118	TX_FQ_CTRL2	TX FIFO Queue Control register 2	0x0000 0000
0x120	TX_FQ_ADD_PT0	TX FIFO Queue 0 Current Address Pointer register	0x0000 0000

0x124	TX_FQ_START_ADD0	TX FIFO Queue 0 Start Address register This register is only accessible in write mode if the TX FIFO Queue 0 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x128	TX_FQ_SIZE0	TX FIFO Queue 0 Size register This register is only accessible in write mode if the TX FIFO Queue 0 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x130	TX_FQ_ADD_PT1	TX FIFO Queue 1 Current Address Pointer register	0x0000 0000
0x134	TX_FQ_START_ADD1	TX FIFO Queue 1 Start Address register This register is only accessible in write mode if the TX FIFO Queue 1 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x138	TX_FQ_SIZE1	TX FIFO Queue 1 Size register This register is only accessible in write mode if the TX FIFO Queue 1 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x140	TX_FQ_ADD_PT2	TX FIFO Queue 2 Current Address Pointer register	0x0000 0000
0x144	TX_FQ_START_ADD2	TX FIFO Queue 2 Start Address register This register is only accessible in write mode if the TX FIFO Queue 2 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000

0x148	TX_FQ_SIZE2	TX FIFO Queue 2 Size register This register is only accessible in write mode if the TX FIFO Queue 2 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x150	TX_FQ_ADD_PT3	TX FIFO Queue 3 Current Address Pointer register	0x0000 0000
0x154	TX_FQ_START_ADD3	TX FIFO Queue 3 Start Address register This register is only accessible in write mode if the TX FIFO Queue 3 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x158	TX_FQ_SIZE3	TX FIFO Queue 3 Size register This register is only accessible in write mode if the TX FIFO Queue 3 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x160	TX_FQ_ADD_PT4	TX FIFO Queue 4 Current Address Pointer register	0x0000 0000
0x164	TX_FQ_START_ADD4	TX FIFO Queue 4 Start Address register This register is only accessible in write mode if the TX FIFO Queue 4 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x168	TX_FQ_SIZE4	TX FIFO Queue 4 Size register This register is only accessible in write mode if the TX FIFO Queue 4 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x170	TX_FQ_ADD_PT5	TX FIFO Queue 5 Current Address Pointer register	0x0000 0000

0x174	TX_FQ_START_ADD5	TX FIFO Queue 5 Start Address register This register is only accessible in write mode if the TX FIFO Queue 5 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x178	TX_FQ_SIZE5	TX FIFO Queue 5 Size register This register is only accessible in write mode if the TX FIFO Queue 5 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x180	TX_FQ_ADD_PT6	TX FIFO Queue 6 Current Address Pointer register	0x0000 0000
0x184	TX_FQ_START_ADD6	TX FIFO Queue 6 Start Address register This register is only accessible in write mode if the TX FIFO Queue 6 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x188	TX_FQ_SIZE6	TX FIFO Queue 6 Size register This register is only accessible in write mode if the TX FIFO Queue 6 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x190	TX_FQ_ADD_PT7	TX FIFO Queue 7 Current Address Pointer register	0x0000 0000
0x194	TX_FQ_START_ADD7	TX FIFO Queue 7 Start Address register This register is only accessible in write mode if the TX FIFO Queue 7 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000

0x198	TX_FQ_SIZE7	TX FIFO Queue 7 Size register This register is only accessible in write mode if the TX FIFO Queue 7 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x300	TX_PQ_STS0	TX Priority Queue Status register	0x0000 0000
0x304	TX_PQ_STS1	TX Priority Queue Status register	0x0000 0000
0x30C	TX_PQ_CTRL0	TX Priority Queue Control register 0	0x0000 0000
0x310	TX_PQ_CTRL1	TX Priority Queue Control register 1 This register is only accessible in write mode if the unlock key sequence has been performed prior to write	0x0000 0000
0x314	TX_PQ_CTRL2	TX Priority Queue Control register 2	0x0000 0000
0x318	TX_PQ_START_ADD	TX Priority Queue Start Address This register is only accessible in write mode if the TX Priority Queue is not busy, see BUSY flag in TX_PQ_STS register. It means TX_PQ_STS register is equal to 0x0. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x400	RX_DESC_ADD_PT	RX descriptor Current Address Pointer	0x0000 0000
0x404	RX_STATISTICS	RX Message Counter register	0x0000 0000
0x408	RX_FQ_STS0	RX FIFO Queue Status register 0	0x0000 0000
0x40C	RX_FQ_STS1	RX FIFO Queue Status register 1	0x0000 0000
0x410	RX_FQ_STS2	RX FIFO Queue Status register 2	0x0000 0000
0x414	RX_FQ_CTRL0	RX FIFO Queue Control register 0	0x0000 0000
0x418	RX_FQ_CTRL1	RX FIFO Queue Control register 1 This register is only accessible in write mode if the unlock key sequence has been performed prior to write	0x0000 0000
0x41C	RX_FQ_CTRL2	RX FIFO Queue Control register 2	0x0000 0000
0x420	RX_FQ_ADD_PT0	RX FIFO Queue 0 Current Address Pointer	0x0000 0000

0x424	RX_FQ_START_ADD0	RX FIFO Queue 0 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x428	RX_FQ_SIZE0	RX FIFO Queue 0 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x42C	RX_FQ_DC_START_ADD0	RX FIFO Queue 0 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x430	RX_FQ_RD_ADD_PT0	RX FIFO Queue 0 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x438	RX_FQ_ADD_PT1	RX FIFO Queue 1 Current Address Pointer	0x0000 0000
0x43C	RX_FQ_START_ADD1	RX FIFO Queue 1 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x440	RX_FQ_SIZE1	RX FIFO Queue 1 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000

0x444	RX_FQ_DC_START_ADD1	RX FIFO Queue 1 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x448	RX_FQ_RD_ADD_PT1	RX FIFO Queue 1 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x450	RX_FQ_ADD_PT2	RX FIFO Queue 2 Current Address Pointer	0x0000 0000
0x454	RX_FQ_START_ADD2	RX FIFO Queue 2 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x458	RX_FQ_SIZE2	RX FIFO Queue 2 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x45C	RX_FQ_DC_START_ADD2	RX FIFO Queue 2 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x460	RX_FQ_RD_ADD_PT2	RX FIFO Queue 2 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x468	RX_FQ_ADD_PT3	RX FIFO Queue 3 Current Address Pointer	0x0000 0000

0x46C	RX_FQ_START_ADD3	RX FIFO Queue 3 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x470	RX_FQ_SIZE3	RX FIFO Queue 3 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x474	RX_FQ_DC_START_ADD3	RX FIFO Queue 3 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x478	RX_FQ_RD_ADD_PT3	RX FIFO Queue 3 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x480	RX_FQ_ADD_PT4	RX FIFO Queue 4 Current Address Pointer	0x0000 0000
0x484	RX_FQ_START_ADD4	RX FIFO Queue 4 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x488	RX_FQ_SIZE4	RX FIFO Queue 4 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000

0x48C	RX_FQ_DC_START_ADD4	RX FIFO Queue 4 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x490	RX_FQ_RD_ADD_PT4	RX FIFO Queue 4 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x498	RX_FQ_ADD_PT5	RX FIFO Queue 5 Current Address Pointer	0x0000 0000
0x49C	RX_FQ_START_ADD5	RX FIFO Queue 5 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x4A0	RX_FQ_SIZE5	RX FIFO Queue 5 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x4A4	RX_FQ_DC_START_ADD5	RX FIFO Queue 5 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x4A8	RX_FQ_RD_ADD_PT5	RX FIFO Queue 5 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x4B0	RX_FQ_ADD_PT6	RX FIFO Queue 6 Current Address Pointer	0x0000 0000

0x4B4	RX_FQ_START_ADD6	RX FIFO Queue 6 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x4B8	RX_FQ_SIZE6	RX FIFO Queue 6 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x4BC	RX_FQ_DC_START_ADD6	RX FIFO Queue 6 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x4C0	RX_FQ_RD_ADD_PT6	RX FIFO Queue 6 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x4C8	RX_FQ_ADD_PT7	RX FIFO Queue 7 Current Address Pointer	0x0000 0000
0x4CC	RX_FQ_START_ADD7	RX FIFO Queue 7 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x4D0	RX_FQ_SIZE7	RX FIFO Queue 7 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000

0x4D4	RX_FQ_DC_START_ADD7	RX FIFO Queue 7 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode	0x0000 0000
0x4D8	RX_FQ_RD_ADD_PT7	RX FIFO Queue 7 Read Address Pointer This register is used only in Continuous Mode.	0x0000 0000
0x600	TX_FILTER_CTRL0	TX Filter Control register 0 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x604	TX_FILTER_CTRL1	TX Filter Control register 1 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x608	TX_FILTER_REFVAL0	TX Filter Reference Value register 0 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x60C	TX_FILTER_REFVAL1	TX Filter Reference Value register 1 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000

0x610	TX_FILTER_REFVAL2	TX Filter Reference Value register 2 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x614	TX_FILTER_REFVAL3	TX Filter Reference Value register 3 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x680	RX_FILTER_CTRL	RX Filter Control register This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x700	TX_FQ_INT_STS	TX FIFO Queue Interrupt Status register	0x0000 0000
0x704	RX_FQ_INT_STS	RX FIFO Queue Interrupt Status register	0x0000 0000
0x708	TX_PQ_INT_STS0	TX Priority Queue Interrupt Status register 0	0x0000 0000
0x70C	TX_PQ_INT_STS1	TX Priority Queue Interrupt Status register 1	0x0000 0000
0x710	STATS_INT_STS	Statistics Interrupt Status register	0x0000 0000
0x714	ERR_INT_STS	Error Interrupt Status register	0x0000 0000
0x718	SFTY_INT_STS	Safety Interrupt Status register	0x0000 0000
0x71C	AXI_ERR_INFO	DMA Error Information	0x0000 0000
0x720	DESC_ERR_INFO0	Descriptor Error Information 0 If the DESC_ERR_INFO0.ADD[31:16] = 0 and DESC_ERR_INFO1.CRC[8:0], DESC_ERR_INFO1.RX_TX and DESC_ERR_INFO1.RC[4:0] are all equal to 0, the faulty descriptor is a TX descriptor fetched from L_MEM	0x0000 0000

0x724	DESC_ERR_INFO1	Descriptor Error Information 1 When the DESC_ERR_INFO1.CRC[8:0], DESC_ERR_INFO1.RX_TX and DESC_ERR_INFO1.RC[4:0] are all equal to 0, the faulty descriptor is a TX descriptor fetched from L_MEM only if the DESC_ERR_INFO0.ADD[31:16] = 0	0x0000 0000
0x728	TX_FILTER_ERR_INFO	TX Filter Error Information	0x0000 0000
0x800	DEBUG_TEST_CTRL	Debug Control register This register is only accessible in write mode if the Test Mode Key sequence has been performed prior to write. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.	0x0000 0000
0x804	INT_TEST0	Interrupt Test register 0 This register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	0x0000 0000
0x808	INT_TEST1	Interrupt Test register 1 This register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	0x0000 0000
0x810	TX_SCAN_FC	TX-SCAN first candidates register This register gives the 4 best candidates evaluated by the TX-Scan. This register gives the first and second highest priority TX descriptor after a TX-Scan. It provides also the third and fourth candidates during a TX-Scan, considering the first and second candidates as already defined by a previous TX-Scan run.	0x0000 0000
0x814	TX_SCAN_BC	TX-SCAN best candidates register This register gives the first and second highest priority TX descriptor after a TX-Scan	0x0000 0000
0x818	TX_FQ_DESC_VALID	Valid TX FIFO Queue descriptors in local memory	0x0000 0000
0x81C	TX_PQ_DESC_VALID	Valid TX Priority Queue descriptors in local memory	0x0000 0000
0x880	CRC_CTRL	CRC Control register	0x0000 0000
0x884	CRC_REG	CRC register	0x0000 0000

Block : xcan_prt_inst_0__reg_axi__900__status

Address	Name	Description	Reset
0x900	ENDN	Endianness Test Register	0x8765 4321
0x904	PREL	PRT Release Identification Register.	0x0540 0000
0x908	STAT	PRT Status Register	0x0000 0010

Block : xcan_prt_inst_0__reg_axi__900__event

Address	Name	Description	Reset
0x920	EVNT	Event Status Flags Register The EVNT Register contains event status flags. The flags are set by the PRT when specific events occur. A software reset clears all flags. A host write access to this register, writing a 1 to a specific flag, clears that flag. When a host write access occurs concurrently with a set condition for a flag, the flag is set.	0x0000 0000

Block : xcan_prt_inst_0__reg_axi__900__control

Address	Name	Description	Reset
0x940	LOCK	Unlock Sequence Register Writing a sequence of specific data words enables the activation of control commands in the registers CTRL and FIMC. Reading this register always gives the value 0x00000000.	0x0000 0000

0x944	CTRL	Control Register Writing to this register controls the CAN protocol operation. Reading this register gives the value 0x00000000. When writing to this register, only one of the four bits TEST, SRES , STRT, or STOP may be written to 1, otherwise the write access takes no effect. The bit IMMD may be written to 1 together with the bit STOP, but not together with one of the other bits.	0x0000 0000
0x948	FIMC	Fault Injection Module Control Register Writing the fault injection position number requires the application of the test mode key sequence before writing to FIMC. This register must be accessed in privileged mode when supported.	0x0000 0000
0x94C	TEST	Hardware Test Functions Register This register is writable after the hardware test mode functions are enabled by writing the test mode key sequence to LOCK and CTRL registers. While the hardware test mode functions are not enabled, this register is read-only. This register must be accessed in privileged mode when supported. The hardware test mode functions are disabled and cleared by the software reset of the PRT.	0x0000 0008

Block : xcan_prt_inst_0__reg_axi__900__configuration

Address	Name	Description	Reset

0x960	MODE	Operating Mode Register Configuration register that is writable while the CAN communication is stopped and that is read-only after the CAN communication is started. This register defines separate operating mode options. The four configuration bits FDOE, XLOE, EFDI, and XLTR, are interrelated according to table Frame Formats defined in Operating Mode chapter.	0x0000 0000
0x964	NBTP	Arbitration Phase Nominal Bit Timing Register Configuration register that is writable while the CAN communication is stopped and that is read-only after the CAN communication is started. This register defines the Nominal Bit Timing as defined in [1].	0x0000 0000
0x968	DBTP	CAN FD Data Phase Bit Timing Register Configuration register that is writable while the CAN communication is stopped and that is read-only after the CAN communication is started. This register defines the FD Data Phase Bit Timing as defined in [1].	0x0000 0000
0x96C	XBTP	CAN XL Data Phase Bit Timing Register Configuration register that is writable while the CAN communication is stopped and is read-only after the CAN communication is started. This register defines the XL Data Phase Bit Timing as defined in [2].	0x0000 0000
0x970	PCFG	PWME Configuration Register Configuration register that is writable while the CAN communication is stopped and is read-only after the CAN communication is started. This register defines the parameters needed for the PWM coding (as described in [2]) in the PWME module for CAN XL transceivers with switchable operating modes	0x0000 0000

Block : xcand_top_irc_inst_0__irc_axi__a00__Event

Address	Name	Description	Reset
0xA00	FUNC_RAW	Functional raw event status register. This register provides information about the occurrence of functional relevant events inside the MH and the PRT. A flag is set when the related event is detected, independent of FUNC_ENA. The flags remain set until the Host CPU clears them by writing a 1 to the corresponding bit position at register FUNC_CLR.	0x0000 0000
0xA04	ERR_RAW	Error raw event status register. This register provides information about the occurrence of functional error relevant events inside the MH and the PRT. A flag is set when the related event is detected, independent of ERR_ENA. The flags remain set until the Host CPU clears them by writing a 1 to the corresponding bit position at register ERR_CLR.	0x0000 0000
0xA08	SAFETY_RAW	Safety raw event status register. This register provides information about the occurrence of safety relevant events inside the MH and the PRT. A flag is set when the related event is detected, independent of SAFETY_ENA. The flags remain set until the Host CPU clears them by writing a 1 to the corresponding bit position at register SAFETY_CLR.	0x0000 0000

Block : xcand_top_irc_inst_0__irc_axi__a00__Control

Address	Name	Description	Reset

0xA10	FUNC_CLR	Functional raw event clear register. Writing a 1 to a certain bit position clears the corresponding bit of register FUNC_RAW. Writing a ’0’ has no effect.	0x0000 0000
0xA14	ERR_CLR	Error raw event clear register. Writing a 1 to a certain bit position clears the corresponding bit of register ERR_RAW. Writing a ’0’ has no effect.	0x0000 0000
0xA18	SAFETY_CLR	Safety raw event clear register. Writing a 1 to a certain bit position clears the corresponding bit of register SAFETY_RAW. Writing a ’0’ has no effect.	0x0000 0000
0xA20	FUNC_ENA	Functional raw event enable register. Any bit in the FUNC_ENA register enables the corresponding bit in the FUNC_RAW to trigger the interrupt line FUNC_INT. The interrupt line gets active high, when at least one RAW/ENA pair is 1, e.g. FUNC_RAW.MH_TX_FQ_IRQ = FUNC_ENA.MH_TX_FQ_IRQ = 1	0x0000 0000
0xA24	ERR_ENA	Error raw event enable register. Any bit in the ERR_ENA register enables the corresponding bit in the ERR_RAW to trigger the interrupt line ERR_INT. The interrupt line gets active high, when at least one RAW/ENA pair is 1, e.g. ERR_RAW.MH_TX_FQ_IRQ = ERR_ENA.MH_TX_FQ_IRQ = 1	0x0000 0000
0xA28	SAFETY_ENA	Safety raw event enable register. Any bit in the SAFETY_ENA register enables the corresponding bit in the SAFETY_RAW to trigger the interrupt line SAFETY_INT. The interrupt line gets active high, when at least one RAW/ENA pair is 1, e.g. SAFETY_RAW.MH_TX_FQ_IRQ = SAFETY_ENA.MH_TX_FQ_IRQ = 1	0x0000 0000

Block : xcand_top_irc_inst_0__irc_axi__a00__Configuration

Address	Name	Description	Reset
0xA30	CAPTURING_MODE	IRC configuration register. This register shows the hardware configuration of the IRC concerning the capturing mode of the event inputs. The IP internal events signals coming from the MH and the PRT require an 'edge sensitive' capturing. That is why the value of this register is 0x7 and cannot be changed.	0x0000 0007

Block : xcand_top_irc_inst_0__irc_axi__a00__Auxiliary

Address	Name	Description	Reset
0xA40	HDP	Hardware Debug Port control register	0x0000 0000

Block: xcand_mh_inst_0__AXI_REG__0__xcand_mh_creg

Description : Global MH control and status registers

Register: VERSION

Description	: Release Identification Register This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x0
Absolute Address	: 0x000
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REL				STEP				SUBSTEP				YEAR			
RO - 0x0				RO - 0x5				RO - 0x6				RO - 0x0			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MON								DAY							
RO - 0x00								RO - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
31:28	REL	Release value, used to identify the main release of the X_CAN	RO - -	0x0
27:24	STEP	Step value according to Release	RO - -	0x5
23:20	SUBSTEP	Sub-Step value according to Step	RO - -	0x6
19:16	YEAR	Define the year of the release using a binary coded decimal representation (0 being 2020 and so forth…). This reset value is defined by the generic parameter DESIGN_TIME_STAMP_G[19:16]. If the generic parameter DESIGN_TIME_STAMP_G is not set, the default value is the one defined here	RO - -	0x0

15:8	MON	Define the month of the release using a binary coded decimal representation (1 being January and so forth). This reset value is defined by the generic parameter DESIGN_TIME_STAMP_G[15:8]. If the generic parameter DESIGN_TIME_STAMP_G is not set, the default value is the one defined here	RO - -	0x00
7:0	DAY	Define the day of the release using a binary coded decimal representation (1 being the first day of the month and so forth). This reset value is defined by the generic parameter DESIGN_TIME_STAMP_G[7:0]. If the generic parameter DESIGN_TIME_STAMP_G is not set, the default value is the one defined here	RO - -	0x00

Register: MH_CTRL

Description : Message Handler Control register
Offset : 0x4
Absolute Address : 0x004
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
															STA RT
															RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset

0	START	Before starting any RX/TX FIFO Queues or TX FIFO Queue slots the MH must write 1 to this bit prior launching the PRT. At initial start, as long as the PRT is not started, this bit can be set back to 0. When set to 1, the global configuration registers are write-protected. As soon as the PRT is started, this bit cannot be set to 0. This bit can only be set to back to 0 if MH_STS.ENABLE = 0 and MH_STS.BUSY =0. For more details on starting/stopping or restarting the MH, refer to the Programming Guidelines chapter.	RW - -	0x0

Register: MH_CFG

Description	: Message Handler Configuration register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x8
Absolute Address	: 0x008
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
													INST_NUM		
													RW - 0x0		
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
					MAX_RETRANS										RX_CONT_DC
					RW - 0x7										RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
18:16	INST_NUM	In case that a cluster of X_CAN is defined, this bit field is used as a unique identifier per instance. This identifier is used by the MH to determine if the TX/RX descriptors are fetched by the right instance, see RX/TX description. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0

10:8	MAX_RETRANS	Maximum number of TX message re-transmissions. Different configurations are possible: 0 -> no re-transmission; 1 to 6 -> 1 to 6 re-transmissions; 7-> unlimited re-transmissions; This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x7
0	RX_CONT_DC	When set to 1, the Continuous mode is active. This mode provides the option to have a linear and continuous memory organization of the RX message data. Only one RX descriptor is used by RX message data and one single data container is required. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0

Register: MH_STS

Description	: Message Handler Status register
Offset	: 0xC
Absolute Address	: 0x00C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
							CLOCK_ACTIVE				ENABLE				BUSY
							RO - 0x0				RO - 0x0				RO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
8	CLOCK_ACTIVE	Status of MH core clock: 0 = clock off, 1 = clock on.	RO - -	0x0
4	ENABLE	Value of the ENABLE signal driven by the PRT. The PRT signalizes via ENABLE whether it is active (ENABLE = 1) and requires message handling or not (ENABLE = 0).	RO - -	0x0

0	BUSY	This bit is the general busy flag, it is an ORED(RX/TX FIFO Queues and TX Priority Queue slots busy flags)	RO - -	0x0

Register: MH_SFTY_CFG

Description	: Message Handler Safety Configuration register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x10
Absolute Address	: 0x010
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRESCALER		PRT_TO_VAL													
RW - 0x0		RW - 0x0000													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MEM_TO_VAL								DMA_TO_VAL							
RW - 0x00								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
31:30	PRESCALER	Prescaler used to generate the timer ticks for the watchdogs. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. According to the value a different clock ratio can be selected: 0: clk divided by 32 1: clk divided by 64 2: clk divided by 128 3: clk divided by 512	RW - -	0x0

29:16	PRT_TO_VAL	This value is used by the watchdog timers for the internal RX_MSG and TX_MSG interfaces. It defines the maximum number of timer ticks until a message has to be transferred from PRT to MH respective MH to PRT. The value must be configured according to the CAN frame which requires the longest time to be transported on the CAN bus. If this value is set to 0 and MH_SFTY_CTRL.PRT_TO_EN = 1 then the DP_TO_ERR interrupt is triggered right away at the beginning of a RX message or when starting a TX message. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0000
15:8	MEM_TO_VAL	This value is used by the watchdog timer for the MEM_AXI interface and defines the maximum number of timer ticks until a read or write access has to be completed. This value must be configured to the expected maximum latency on the MEM_AXI interface. If this value is set to 0 and MH_SFTY_CTRL.MEM_TO_EN = 1 then the MEM_TO_ERR interrupt is triggered right away when accessing the L_MEM. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x00
7:0	DMA_TO_VAL	This value is used by the watchdog timer for the DMA_AXI interface and defines the maximum number of timer ticks until a read or write access has to be completed. This value must be configured according to the maximum system latency, expected on the DMA_AXI interface. If this value is set to 0 and MH_SFTY_CTRL.DMA_TO_EN = 1 then the DMA_TO_ERR interrupt is triggered right away when accessing the S_MEM. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x00

Register: MH_SFTY_CTRL

Description	: Message Handler Safety Control register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x14
Absolute Address	: 0x014

Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					PRT_TO_EN	MEM_TO_EN	DMA_TO_EN	DMA_CH_CHK_EN	RX_AP_PARITY_EN	TX_AP_PARITY_EN	TX_DP_PARITY_EN	RX_DP_PARITY_EN	MEM_PROT_EN	RX_DESC_CRC_EN	TX_DESC_CRC_EN
					RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
10	PRT_TO_EN	When set to 1, the watchdogs for the internal RX_MSG and TX_MSG interfaces are enabled, otherwise disabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
9	MEM_TO_EN	When set to 1, the watchdog for the MEM_AXI interface is enabled, otherwise disabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
8	DMA_TO_EN	When set to 1, the watchdog for the DMA_AXI interface is enabled, otherwise disabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
7	DMA_CH_CHK_EN	When set to 1, the read/write DMA channels routing is checked. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
6	RX_AP_PARITY_EN	When set to 1, the address pointer parity check on the RX path is enabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
5	TX_AP_PARITY_EN	When set to 1, the address pointer parity check on the TX path is enabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
4	TX_DP_PARITY_EN	When set to 1, the data path parity check performed on the TX path is enabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0

3	RX_DP_PARITY_EN	When set to 1, the data path parity check performed on the RX path is enabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
2	MEM_PROT_EN	When set to 1, the sfty_err signal from the local memory interface is checked. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
1	RX_DESC_CRC_EN	When set to 1, the CRC check for the RX descriptors is enabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
0	TX_DESC_CRC_EN	When set to 1, the CRC check for the TX descriptors is enabled. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0

Register: RX_FILTER_MEM_ADD

Description	: RX Filter Base Address register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x18
Absolute Address	: 0x018
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE_ADDR															
RW - 0x0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

15:0	BASE_ADDR	Define the base address where the RX filter elements are defined in L_MEM (up to 64Kbytes can be addressed). The BASE_ADDR[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0000

Register: TX_DESC_MEM_ADD

Description	: TX Descriptor Base Address register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x1C
Absolute Address	: 0x01C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PQ_BASE_ADDR															
RW - 0x0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
FQ_BASE_ADDR															
RW - 0x0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:16	PQ_BASE_ADDR	Define the base address where the TX Priority Queue descriptors are stored in L_MEM (up to 64Kbytes can be addressed). The PQ_BASE_ADDR[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0000

15:0	FQ_BASE_ADDR	Define the base address where the TX FIFO Queue descriptors are stored in L_MEM (up to 64Kbytes can be addressed). The FQ_BASE_ADDR[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0000

Register: AXI_ADD_EXT

Description	: AXI address extension register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x20
Absolute Address	: 0x020
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the MSB of the read/write AXI address bus used on the DMA_AXI interface. If not required, leave the default value and do not connect the upper part of the DMA_AXI read/write address bus. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0000 0000

Register: AXI_PARAMS

Description	: AXI parameter register This register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x24
Absolute Address	: 0x024
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
										AW_MAX_PEND				AR_MAX_PEND	
										RW - 0x0				RW - 0x0	

Bit	Name	Description	SW Access HW Access Protection	Reset
5:4	AW_MAX_PEND	AW_MAX_PEND[1:0] defines the maximum write pending transactions on DMA_AXI interface: 0 -> no write transfer; 1 -> 1 outstanding write transaction allowed; 2 -> 2 outstanding write transactions, 3 -> 3 outstanding write transactions. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0
1:0	AR_MAX_PEND	AR_MAX_PEND[1:0] defines the maximum read pending transactions on DMA_AXI interface: 0 -> no read transfer; 1 -> 1 outstanding read transaction; 2 -> 2 outstanding read transactions, 3 -> 3 outstanding read transactions. This bit field register is only accessible in write mode if the MH is not started, see MH_CTRL.START = 0.	RW - -	0x0

Register: MH_LOCK

Description	: Message Handler Lock register
Offset	: 0x28
Absolute Address	: 0x028
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

TMK															
RW - 0x0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ULK															
RW - 0x0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:16	TMK	Test mode key register. Two consecutive writes to this bit field, starting with 0x6789 and 0x9876, must be done before writing to the DEBUG_TEST_CTRL register.	RW - -	0x0000
15:0	ULK	Unlock key register. Two consecutive writes to this bit field, starting with 0x1234 and 0x04321, must be done before writing to a register being locked.	RW - -	0x0000

Register: TX_DESC_ADD_PT

Description	: TX descriptor current address pointer register
Offset	: 0x100
Absolute Address	: 0x100
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Address used to fetch a TX descriptor for the TX FIFO Queues or TX Priority Queue slots. It could be for several reasons: a new message needs to be fetched from a TX FIFO Queue or a new message is defined in a TX Priority Queue slot. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_STATISTICS

Description : TX Message Counter register
Offset : 0x104
Absolute Address : 0x104
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				UNSUCC											
				RW - 0x000											
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
				SUCC											
				RW - 0x000											

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	UNSUCC	Counter incremented with every unsuccessful transmission of a CAN message to the CAN bus. The counter wraps automatically to 0 and can be cleared when writing 0 to the bit field. A STATS_IRQ interrupt is generated when the counter wraps.	RW - -	0x000
11:0	SUCC	Counter incremented with every successful transmission of a CAN message to the CAN bus. The counter wraps automatically to 0 and can be cleared when writing 0 to the bit field. A STATS_IRQ interrupt is generated when the counter wraps.	RW - -	0x000

Register: TX_FQ_STS0

Description : TX FIFO Queue Status register
Offset : 0x108
Absolute Address : 0x108
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								STOP							
								RO - 0x00							
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								BUSY							
								RO - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
23:16	STOP	When STOP[n] = 1 the TX FIFO Queue n is on hold, this means the FIFO Queue is started and running but waits for the SW to keep going. The STOP[n] can be set only if the BUSY[n] = 1. Several root causes may lead to this state: an error is detected, or a TX descriptor is not valid. To identify the potential issues, refer to the TX_FQ_STS1 register. In order to keep going with the TX FIFO Queue n, a write to the TX_FQ_CTRL0.START[n] is required. When BUSY[n] = 0, this bit is automatically set to 0	RO - -	0x00
7:0	BUSY	When BUSY[n] = 1 the TX FIFO Queue n is active, this means the FIFO Queue is started and running (TX message defined in the TX FIFO Queue n can be processed). When the BUSY[n] = 0, the TX FIFO Queue n is stopped and would require a write to the TX_FQ_CTRL0.START[n] to make it active again. A TX FIFO Queue can go inactive if the END bit in the last TX descriptor of a TX message is set. In this case the, the BUSY[n] = 0 can occur only if the TX header descriptor of this last message has been acknowledged for the TX FIFO Queue n. When the TX FIFO Queue n is aborted, the BUSY[n] flag is set to 0 only when no acknowledge is pending.	RO - -	0x00

Register: TX_FQ_STS1

Description	: TX FIFO Queue Status register
Offset	: 0x10C
Absolute Address	: 0x10C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								ERROR							
								RO - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								UNVALID							
								RO - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
23:16	ERROR	When ERROR[n] = 1 the TX FIFO Queue n is on hold due to an inconsistent TX descriptor was loaded, see chapter Descriptor Protection.	RO - -	0x00
7:0	UNVALID	When UNVALID[n] = 1 the TX FIFO Queue n is on hold due to an TX descriptor with VALID=0 was loaded.	RO - -	0x00

Register: TX_FQ_CTRL0

Description	: TX FIFO Queue Control register 0
Offset	: 0x110
Absolute Address	: 0x110
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								START							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
7:0	START	When writing a 1 to the START[n], the TX FIFO Queue n is started. This bit is autocleared. Once started, the TX_FQ_STS0.BUSY[n] is set to 1. The MH must be started prior to any TX FIFO Queue start (MH_STS.BUSY set to 1). A TX FIFO Queue n can only be started if TX_FQ_CTRL2.ENABLE[n] is set to 1 and in order to avoid a dead lock situation with the PRT, the ENABLE signal from the PRT is high.	RW - -	0x00

Register: TX_FQ_CTRL1

Description	: TX FIFO Queue Control register 1 This register is only accessible in write mode if the unlock key sequence has been performed prior to write

Offset : 0x114
Absolute Address : 0x114
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								ABORT							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
7:0	ABORT	When ABORT[n] is set to 1, the TX FIFO Queue n is aborted. Once set to 1, the MH will abort all pending transaction related to the TX FIFO Queue n whenever required. This bit must be set back to 0 only when the TX FIFO Queue n is inactive, TX_FQ_STS0.BUSY[n] = 0. This bit field register is only accessible in write mode if the unlock key sequence has been performed prior to write.	RW - -	0x00

Register: TX_FQ_CTRL2

Description : TX FIFO Queue Control register 2
Offset : 0x118
Absolute Address : 0x118
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								ENABLE							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset

7:0	ENABLE	When ENABLE[n] is set to 1, the TX FIFO Queue n is enabled. A TX FIFO Queue cannot be started if it is not enabled. Aborting a not started TX FIFO Queue has no effect.	RW - -	0x00

Register: TX_FQ_ADD_PT0

Description : TX FIFO Queue 0 Current Address Pointer register
Offset : 0x120
Absolute Address : 0x120
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD0

Description : TX FIFO Queue 0 Start Address register
This register is only accessible in write mode if the TX FIFO Queue 0 is not busy, see BUSY flag in TX_FQ_STS0 register.
This register is protected by a register bank CRC defined in CRC_REG register.
Offset : 0x124
Absolute Address : 0x124
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															

RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 0 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE0

Description	: TX FIFO Queue 0 Size register This register is only accessible in write mode if the TX FIFO Queue 0 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x128
Absolute Address	: 0x128
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset

9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 0 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_FQ_ADD_PT1

Description : TX FIFO Queue 1 Current Address Pointer register
Offset : 0x130
Absolute Address : 0x130
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD1

Description	: TX FIFO Queue 1 Start Address register This register is only accessible in write mode if the TX FIFO Queue 1 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x134
Absolute Address	: 0x134
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 1 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE1

Description	: TX FIFO Queue 1 Size register This register is only accessible in write mode if the TX FIFO Queue 1 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x138
Absolute Address	: 0x138
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

	MAX_DESC
	RW - 0x000

Bit	Name	Description	SW Access HW Access Protection	Reset
9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 1 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_FQ_ADD_PT2

Description : TX FIFO Queue 2 Current Address Pointer register
Offset : 0x140
Absolute Address : 0x140
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD2

Description	: TX FIFO Queue 2 Start Address register This register is only accessible in write mode if the TX FIFO Queue 2 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x144
Absolute Address	: 0x144
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 2 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE2

Description	: TX FIFO Queue 2 Size register This register is only accessible in write mode if the TX FIFO Queue 2 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x148
Absolute Address	: 0x148
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset
9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 2 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_FQ_ADD_PT3

Description : TX FIFO Queue 3 Current Address Pointer register
Offset : 0x150
Absolute Address : 0x150
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD3

Description	: TX FIFO Queue 3 Start Address register This register is only accessible in write mode if the TX FIFO Queue 3 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x154
Absolute Address	: 0x154
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 3 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE3

Description	: TX FIFO Queue 3 Size register This register is only accessible in write mode if the TX FIFO Queue 3 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.

Offset : 0x158
Absolute Address : 0x158
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset
9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 3 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_FQ_ADD_PT4

Description : TX FIFO Queue 4 Current Address Pointer register
Offset : 0x160
Absolute Address : 0x160
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD4

Description	: TX FIFO Queue 4 Start Address register This register is only accessible in write mode if the TX FIFO Queue 4 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x164
Absolute Address	: 0x164
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 4 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE4

Description	: TX FIFO Queue 4 Size register This register is only accessible in write mode if the TX FIFO Queue 4 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x168
Absolute Address	: 0x168
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset
9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 4 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_FQ_ADD_PT5

Description	: TX FIFO Queue 5 Current Address Pointer register
Offset	: 0x170
Absolute Address	: 0x170
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

VAL
RO - 0x0000 0000

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD5

Description	: TX FIFO Queue 5 Start Address register This register is only accessible in write mode if the TX FIFO Queue 5 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x174
Absolute Address	: 0x174
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 5 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE5

Description	: TX FIFO Queue 5 Size register This register is only accessible in write mode if the TX FIFO Queue 5 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x178
Absolute Address	: 0x178
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset
9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 5 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_FQ_ADD_PT6

Description	: TX FIFO Queue 6 Current Address Pointer register
Offset	: 0x180
Absolute Address	: 0x180
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD6

Description	: TX FIFO Queue 6 Start Address register This register is only accessible in write mode if the TX FIFO Queue 6 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x184
Absolute Address	: 0x184
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 6 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE6

Description	: TX FIFO Queue 6 Size register This register is only accessible in write mode if the TX FIFO Queue 6 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x188
Absolute Address	: 0x188
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset
9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 6 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_FQ_ADD_PT7

Description : TX FIFO Queue 7 Current Address Pointer register
Offset : 0x190
Absolute Address : 0x190
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the header descriptor address of the TX message being in used by the arbiter for the TX FIFO Queue. To follow TX descriptors over time while running TX FIFO Queues, refer to the TX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: TX_FQ_START_ADD7

Description : TX FIFO Queue 7 Start Address register
This register is only accessible in write mode if the TX FIFO Queue 7 is not busy, see BUSY flag in TX_FQ_STS0 register.
This register is protected by a register bank CRC defined in CRC_REG register.
Offset : 0x194
Absolute Address : 0x194
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the TX FIFO Queue link list descriptor in the system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX FIFO Queue 7 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x0000 0000

Register: TX_FQ_SIZE7

Description	: TX FIFO Queue 7 Size register This register is only accessible in write mode if the TX FIFO Queue 7 is not busy, see BUSY flag in TX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x198
Absolute Address	: 0x198
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset

9:0	MAX_DESC	Define the maximum number of TX descriptors in the TX FIFO Queue link list descriptors. It is important to note that MAX_DESC = 0 does not prevent the TX FIFO Queue to be enabled and started. An active and running TX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no TX descriptor is defined. The memory size to allocate is MAX_DESC * 32bytes for MAX_DESC > = 1. This bit field register is only accessible in write mode if the TX FIFO Queue 7 is not busy, see BUSY flag in TX_FQ_STS0 register.	RW - -	0x000

Register: TX_PQ_STS0

Description	: TX Priority Queue Status register
Offset	: 0x300
Absolute Address	: 0x300
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BUSY															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BUSY															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	BUSY	When BUSY[n] = 1, the TX Priority Queue slot n is busy, which means that the TX descriptor in the slot n is being loaded in L_MEM and considered by the TX-Scan. As long as this bit remains high, the message attached to the slot n has not been sent yet. The BUSY[n] = 0 can occur only if the TX header descriptor of the slot n has been acknowledged.	RO - -	0x0000 0000

Register: TX_PQ_STS1

Description	: TX Priority Queue Status register

Offset : 0x304
Absolute Address : 0x304
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SENT															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SENT															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	SENT	When SENT[n] = 1 the TX message assigned to the TX Priority Queue slot n has been transmitted and the TX descriptor attached to the slot n is acknowledged. This bit will be cleared once a new start on this slot will occur.	RO - -	0x0000 0000

Register: TX_PQ_CTRL0

Description : TX Priority Queue Control register 0
Offset : 0x30C
Absolute Address : 0x30C
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
START															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
START															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	START	When writing a 1 to the START[n], the TX Priority Queue slot n is started and running. This bit is autocleared and once started, the TX_PQ_STS0.BUSY[n] is set to 1. The MH must be started prior to any TX Priority Queue slot start (MH_STS.BUSY set to 1). A TX Priority Queue slot n can only be started if TX_PQ_CTRL2.ENABLE[n] is set to 1 and in order to avoid a dead lock situation with the PRT, the ENABLE signal from the PRT is high.	RW - -	0x0000 0000

Register: TX_PQ_CTRL1

Description	: TX Priority Queue Control register 1 This register is only accessible in write mode if the unlock key sequence has been performed prior to write
Offset	: 0x310
Absolute Address	: 0x310
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ABORT															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABORT															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	ABORT	When ABORT[n] is set to 1, the TX Priority Queue slot n is aborted. This bit must be set back to 0 only when the TX Priority Queue slot n is inactive, TX_FQ_STS0.BUSY[n] = 0. A TX message attached to a slot can only be aborted if it is not stored in the two internal buffers holding the two best candidates for the next TX message. Despite a TX message is aborted, it may have been sent, check the TX_PQ_STS1.SENT[n] bit register for the slot n. This bit field register is only accessible in write mode if the unlock key sequence has been performed prior to write.	RW - -	0x0000 0000

Register: TX_PQ_CTRL2

Description	: TX Priority Queue Control register 2
Offset	: 0x314
Absolute Address	: 0x314
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ENABLE															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ENABLE															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	ENABLE	When ENABLE[n] is set to 1, the slot n in the TX Priority Queue is enabled. A TX Priority Queue slot cannot be started if not enabled. Aborting a not started slot n has no effect	RW - -	0x0000 0000

Register: TX_PQ_START_ADD

Description	: TX Priority Queue Start Address This register is only accessible in write mode if the TX Priority Queue is not busy, see BUSY flag in TX_PQ_STS register. It means TX_PQ_STS register is equal to 0x0. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x318
Absolute Address	: 0x318
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	Define the start address of the TX Priority Queue in the system memory. All TX header descriptors in the TX Priority Queue are continuously defined from this start address. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This bit field register is only accessible in write mode if the TX Priority Queue is not busy, see BUSY flag in TX_PQ_STS register	RW - -	0x0000 0000

Register: RX_DESC_ADD_PT

Description	: RX descriptor Current Address Pointer
Offset	: 0x400
Absolute Address	: 0x400
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the address used to fetch the current RX descriptor. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_STATISTICS

Description	: RX Message Counter register
Offset	: 0x404
Absolute Address	: 0x404
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				UNSUCC											
				RW - 0x000											

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				SUCC											
				RW - 0x000											

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	UNSUCC	Counter incremented with every unsuccessful reception of a CAN message from the CAN bus. The counter wraps automatically to 0 and can be cleared when writing 0x00 to the bit field. . An interrupt is generated when the counter wraps.	RW - -	0x000
11:0	SUCC	Counter incremented with every successful reception of a CAN message from the CAN bus. The counter wraps automatically to 0 and can be cleared when writing 0x00 to the bit field. An interrupt is generated when the counter wraps.	RW - -	0x000

Register: RX_FQ_STS0

Description	: RX FIFO Queue Status register 0
Offset	: 0x408
Absolute Address	: 0x408
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								STOP							
								RO - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								BUSY							
								RO - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset

23:16	STOP	When STOP[n] = 1 the RX FIFO Queue n is on hold, it means started but waiting for the SW to react. The STOP[n] can be set only if the BUSY[n] = 1. Several root causes may lead to the RX FIFO Queue n to stop: an error is detected, or an RX descriptor is not valid, or the FIFO is full. To identify the potential issues, refer to the RX_FQ_STS1 and RX_FQ_STS2 registers. In order to keep going with the RX FIFO Queue n, a write to the RX_FQ_CTRL0.START[n] is required. When BUSY[n] = 0, this bit is automatically set to 0	RO - -	0x00
7:0	BUSY	When BUSY[n] = 1 the RX FIFO Queue n is busy, this means the FIFO Queue is started and running (RX message to be written to the RX FIFO Queue can be processed). When the BUSY[n] = 0, the RX FIFO Queue n is stopped and would require a write to the RX_FQ_CTRL0.START[n] to make it active again. When the RX FIFO Queue n is aborted, the BUSY[n] flag is set to 0 only when no acknowledge is pending.	RO - -	0x00

Register: RX_FQ_STS1

Description : RX FIFO Queue Status register 1
Offset : 0x40C
Absolute Address : 0x40C
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								ERROR							
								RO - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								UNVALID							
								RO - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
23:16	ERROR	When ERROR[n] = 1 the RX FIFO Queue n is on hold due to an inconsistent RX descriptor being loaded, see chapter Descriptor Protection.	RO - -	0x00

7:0	UNVALID	When UNVALID[n] = 1 the RX FIFO Queue n is on hold due to an RX descriptor detected with VALID=0	RO - -	0x00

Register: RX_FQ_STS2

Description : RX FIFO Queue Status register 2
Offset : 0x410
Absolute Address : 0x410
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								DC_FULL							
								RO - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
7:0	DC_FULL	When DC_FULL[n] = 1 the RX FIFO Queue n is stopped due to the RX FIFO Queue n being full. This register is relevant only for the Continuous Mode as in Normal mode, there is no need to provide such information to the MH	RO - -	0x00

Register: RX_FQ_CTRL0

Description : RX FIFO Queue Control register 0
Offset : 0x414
Absolute Address : 0x414
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								START							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
7:0	START	When writing a 1 to the START[n], the RX FIFO Queue n is started. This bit is autocleared and once started, the RX_FQ_STS0.BUSY[n] is set to 1. The MH must be started prior to any RX FIFO Queue start (MH_STS.BUSY set to 1). An RX FIFO Queue n can only be started if RX_FQ_CTRL2.ENABLE[n] is set to 1.	RW - -	0x00

Register: RX_FQ_CTRL1

Description	: RX FIFO Queue Control register 1 This register is only accessible in write mode if the unlock key sequence has been performed prior to write
Offset	: 0x418
Absolute Address	: 0x418
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								ABORT							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
7:0	ABORT	When ABORT[n] is set to 1, the RX FIFO Queue n is aborted. Once set to 1, the MH will abort all pending transactions related to the RX FIFO Queue n whenever required. The abort can be effective only if the RX FIFO Queue n is enabled. This bit must be set back to 0 only when the RX FIFO Queue n is inactive, RX_FQ_STS0.BUSY[n] = 0. This bit field register is only accessible in write mode if the unlock key sequence has been performed prior to write.	RW - -	0x00

Register: RX_FQ_CTRL2

Description : RX FIFO Queue Control register 2
Offset : 0x41C
Absolute Address : 0x41C
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
								ENABLE							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
7:0	ENABLE	When ENABLE[n] is set to 1, the RX FIFO Queue n is enabled. The RX FIFO Queue n cannot be started if not enabled. The abort of an RX FIFO Queue n not started would have no effect.	RW - -	0x00

Register: RX_FQ_ADD_PT0

Description : RX FIFO Queue 0 Current Address Pointer
Offset : 0x420
Absolute Address : 0x420
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 0 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD0

Description	: RX FIFO Queue 0 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x424
Absolute Address	: 0x424
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE0

Description	: RX FIFO Queue 0 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x428

Absolute Address : 0x428
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD0

Description	: RX FIFO Queue 0 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x42C
Absolute Address	: 0x42C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 0 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT0

Description	: RX FIFO Queue 0 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x430
Absolute Address	: 0x430
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure that enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT0 register to be equal to the RX_FQ_START_ADDR0 registers. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: RX_FQ_ADD_PT1

Description	: RX FIFO Queue 1 Current Address Pointer
Offset	: 0x438
Absolute Address	: 0x438
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 1 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD1

Description	: RX FIFO Queue 1 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x43C
Absolute Address	: 0x43C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE1

Description	: RX FIFO Queue 1 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x440
Absolute Address	: 0x440
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									
						RW - 0x000									

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD1

Description	: RX FIFO Queue 1 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x444
Absolute Address	: 0x444
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 1 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT1

Description	: RX FIFO Queue 1 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x448
Absolute Address	: 0x448
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT1 register to be equal to the RX_FQ_START_ADDR1 registers. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: RX_FQ_ADD_PT2

Description	: RX FIFO Queue 2 Current Address Pointer
Offset	: 0x450
Absolute Address	: 0x450
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 2 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD2

Description	: RX FIFO Queue 2 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x454
Absolute Address	: 0x454
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE2

Description	: RX FIFO Queue 2 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x458
Absolute Address	: 0x458
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									

	RW - 0x000

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD2

Description	: RX FIFO Queue 2 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x45C
Absolute Address	: 0x45C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 2 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT2

Description	: RX FIFO Queue 2 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x460
Absolute Address	: 0x460
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT2 register to be equal to the RX_FQ_START_ADDR2 registers. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: RX_FQ_ADD_PT3

Description : RX FIFO Queue 3 Current Address Pointer
Offset : 0x468
Absolute Address : 0x468
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 3 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD3

Description	: RX FIFO Queue 3 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x46C
Absolute Address	: 0x46C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE3

Description	: RX FIFO Queue 3 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x470
Absolute Address	: 0x470
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									

	RW - 0x000

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD3

Description	: RX FIFO Queue 3 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x474
Absolute Address	: 0x474
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 3 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT3

Description	: RX FIFO Queue 3 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x478
Absolute Address	: 0x478
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT3 register to be equal to the RX_FQ_START_ADDR3 registers. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: RX_FQ_ADD_PT4

Description	: RX FIFO Queue 4 Current Address Pointer
Offset	: 0x480
Absolute Address	: 0x480
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 4 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD4

Description	: RX FIFO Queue 4 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x484
Absolute Address	: 0x484
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE4

Description	: RX FIFO Queue 4 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x488
Absolute Address	: 0x488
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
						MAX_DESC									

	RW - 0x000

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD4

Description	: RX FIFO Queue 4 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x48C
Absolute Address	: 0x48C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 4 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT4

Description	: RX FIFO Queue 4 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x490
Absolute Address	: 0x490
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT4 register to be equal to the RX_FQ_START_ADDR4 registers. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: RX_FQ_ADD_PT5

Description	: RX FIFO Queue 5 Current Address Pointer
Offset	: 0x498
Absolute Address	: 0x498
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 5 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD5

Description	: RX FIFO Queue 5 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x49C
Absolute Address	: 0x49C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE5

Description	: RX FIFO Queue 5 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x4A0
Absolute Address	: 0x4A0
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
						MAX_DESC									

	RW - 0x000

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD5

Description	: RX FIFO Queue 5 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x4A4
Absolute Address	: 0x4A4
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 5 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT5

Description	: RX FIFO Queue 5 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x4A8
Absolute Address	: 0x4A8
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT5 register to be equal to the RX_FQ_START_ADDR5 registers. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: RX_FQ_ADD_PT6

Description	: RX FIFO Queue 6 Current Address Pointer
Offset	: 0x4B0
Absolute Address	: 0x4B0
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 6 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD6

Description	: RX FIFO Queue 6 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x4B4
Absolute Address	: 0x4B4
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE6

Description	: RX FIFO Queue 6 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x4B8
Absolute Address	: 0x4B8
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									

	RW - 0x000

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD6

Description	: RX FIFO Queue 6 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x4BC
Absolute Address	: 0x4BC
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 6 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT6

Description	: RX FIFO Queue 6 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x4C0
Absolute Address	: 0x4C0
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT6 register to be equal to the RX_FQ_START_ADDR6 registers. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: RX_FQ_ADD_PT7

Description : RX FIFO Queue 7 Current Address Pointer
Offset : 0x4C8
Absolute Address : 0x4C8
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RO - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Provide the current RX Header Descriptor address pointer for the RX FIFO Queue 7 in the system memory. To follow RX descriptor over time, refer to the RX_DESC_ADD_PT register. This address value is always word aligned (32bit).	RO - -	0x0000 0000

Register: RX_FQ_START_ADD7

Description	: RX FIFO Queue 7 link list Start Address This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x4CC
Absolute Address	: 0x4CC
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the start address of the RX FIFO Queue link list descriptor in system memory. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_SIZE7

Description	: RX FIFO Queue 7 link list and data container Size This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x4D0
Absolute Address	: 0x4D0
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				DC_SIZE											
				RW - 0x000											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
						MAX_DESC									

	RW - 0x000

Bit	Name	Description	SW Access HW Access Protection	Reset
27:16	DC_SIZE	In Normal mode only the DC_SIZE[6:0] is used to define the maximum size of an RX data container for the RX FIFO Queue. The data container size is DC_SIZE[6:0] * 32bytes and one is attached to every RX descriptor. In continuous mode, it defines the size of the single data container used to write all RX messages. The overall data container size is DC_SIZE[11:0] * 32bytes for MAX_DESC > = 1. When set to 0, the RX FIFO Queue can be enabled but not started. This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000
9:0	MAX_DESC	Define the maximum number of descriptors in the RX FIFO Queue link list. It is important to note that MAX_DESC = 0 does not prevent the RX FIFO Queue to be enabled and started. An active and running RX FIFO Queue with MAX_DESC = 0 is not allowed and will result in a DESC_ERR interrupt if no RX descriptor is defined. The size to be allocated to the link list must be equal to MAX_DESC * 16bytes for MAX_DESC >= 1. This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x000

Register: RX_FQ_DC_START_ADD7

Description	: RX FIFO Queue 7 Data Container Start Address This register is accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register. This register is protected by a register bank CRC defined in CRC_REG register. This register is used only in Continuous Mode
Offset	: 0x4D4
Absolute Address	: 0x4D4
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	Define the Data Container Start Address in system memory. This bit field is relevant only when the MH is configured in Continuous Mode. The VAL[1:0] bits are always assumed to be 0b00 whatever the value written. This address value must always be word aligned (32bit). This register is only accessible in write mode if the RX FIFO Queue 7 is not busy, see BUSY flag in RX_FQ_STS0 register	RW - -	0x0000 0000

Register: RX_FQ_RD_ADD_PT7

Description	: RX FIFO Queue 7 Read Address Pointer This register is used only in Continuous Mode.
Offset	: 0x4D8
Absolute Address	: 0x4D8
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset

31:0	VAL	The SW uses this register to indicate the Data Read Address of the RX message being read to the MH. This address must point to the last word of the RX message considered in the data container. This bit field is relevant only when the MH is configured in Continuous mode. The MH uses this information to ensure enough memory space is available to write the next message. For an initial start, it is mandatory to set VAL[1:0] to 0b11, to avoid RX_FQ_RD_ADD_PT7 register to be equal to the RX_FQ_START_ADDR7 register. Excepted for the initial value, the address value must always be word aligned (32bit), VAL[1:0] must be set to 0b00.	RW - -	0x0000 0000

Register: TX_FILTER_CTRL0

Description	: TX Filter Control register 0 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x600
Absolute Address	: 0x600
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
											IRQ_EN	EN	CC_CAN	CAN_FD	MODE
											RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MASK								COMB							
RW - 0x00								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
20	IRQ_EN	When set to 1, enable the interrupt tx_filter_irq to be triggered. The interrupt is triggered when a message is rejected. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0

19	EN	When set to 1, enable the TX filter for all TX message to be sent. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0
18	CC_CAN	When set to 1 reject Classic CAN messages, otherwise accept them. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0
17	CAN_FD	When set to 1 reject CAN-FD messages, otherwise accept them. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0
16	MODE	When set to 1 accept on match, otherwise reject on match. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0
15:8	MASK	When MASK[n] =1 the reference values REF_VAL0/1 or REF_VAL2/3 are combined to define a value (REF_VAL0 or REF_VAL2) and a mask (REF_VAL1 or REF_VAL3). Otherwise, the comparison uses the REF_VAL0/1/2/3 bit field as reference value only. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
7:0	COMB	When COMB[n] =1 the comparison attached to the reference values (REF_VAL0 and REF_VAL1) or (REF_VAL2 and REF_VAL3) are required to accept a TX message. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00

Register: TX_FILTER_CTRL1

Description	: TX Filter Control register 1 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x604
Absolute Address	: 0x604
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FIELD															
RW - 0x0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

VALID
RW - 0x0000

Bit	Name	Description	SW Access HW Access Protection	Reset
31:16	FIELD	When FIELD[n] = 1 the TX filter element n is considering SDT, otherwise VCID, to compare with the reference value defined in TX_FILTER_REFVAL0/1/2/3. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The reference value to be set is defined as follow: FIELD[n] is assigned to TX_FILTER_REFVAL0.REF_VAL{n} (n € {0, 1, 2, 3}) FIELD[n+4] is assigned to TX_FILTER_REFVAL1.REF_VAL{n} (n € {0, 1, 2, 3}) FIELD[n+8] is assigned to TX_FILTER_REFVAL2.REF_VAL{n} (n € {0, 1, 2, 3}) FIELD[n+12] is assigned to TX_FILTER_REFVAL3.REF_VAL{n} (n € {0, 1, 2, 3})	RW - -	0x0000
15:0	VALID	When VALID[n] = 1 the reference value defined for the TX filter n is valid. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The valid reference value used is defined as follow: VALID[n] is assigned to TX_FILTER_REFVAL0.REF_VAL{n} (n € {0, 1, 2, 3}) VALID[n+4] is assigned to TX_FILTER_REFVAL1.REF_VAL{n} (n € {0, 1, 2, 3}) VALID[n+8] is assigned to TX_FILTER_REFVAL2.REF_VAL{n} (n € {0, 1, 2, 3}) VALID[n+12] is assigned to TX_FILTER_REFVAL3.REF_VAL{n} (n € {0, 1, 2, 3})	RW - -	0x0000

Register: TX_FILTER_REFVAL0

Description	: TX Filter Reference Value register 0 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x608
Absolute Address	: 0x608
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REF_VAL3								REF_VAL2							
RW - 0x00								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REF_VAL1								REF_VAL0							
RW - 0x00								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
31:24	REF_VAL3	Define the reference value 3. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
23:16	REF_VAL2	Define the reference value 2. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
15:8	REF_VAL1	Define the reference value 1. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
7:0	REF_VAL0	Define the reference value 0. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00

Register: TX_FILTER_REFVAL1

Description	: TX Filter Reference Value register 1 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x60C
Absolute Address	: 0x60C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REF_VAL3								REF_VAL2							
RW - 0x00								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REF_VAL1								REF_VAL0							
RW - 0x00								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
31:24	REF_VAL3	Define the reference value 3. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
23:16	REF_VAL2	Define the reference value 2. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
15:8	REF_VAL1	Define the reference value 1. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
7:0	REF_VAL0	Define the reference value 0. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00

Register: TX_FILTER_REFVAL2

Description	: TX Filter Reference Value register 2 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x610
Absolute Address	: 0x610
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REF_VAL3								REF_VAL2							
RW - 0x00								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REF_VAL1								REF_VAL0							

RW - 0x00	RW - 0x00

Bit	Name	Description	SW Access HW Access Protection	Reset
31:24	REF_VAL3	Define the reference value 3. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
23:16	REF_VAL2	Define the reference value 2. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
15:8	REF_VAL1	Define the reference value 1. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
7:0	REF_VAL0	Define the reference value 0. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00

Register: TX_FILTER_REFVAL3

Description	: TX Filter Reference Value register 3 This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x614
Absolute Address	: 0x614
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REF_VAL3								REF_VAL2							
RW - 0x00								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REF_VAL1								REF_VAL0							
RW - 0x00								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset

31:24	REF_VAL3	Define the reference value 3. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
23:16	REF_VAL2	Define the reference value 2. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
15:8	REF_VAL1	Define the reference value 1. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
7:0	REF_VAL0	Define the reference value 0. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00

Register: RX_FILTER_CTRL

Description	: RX Filter Control register This register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x680
Absolute Address	: 0x680
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
										ANFF	ANMF		ANMF_FQ		
										RW - 0x0	RW - 0x0		RW - 0x0		
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
			THRESHOLD					NB_FE							
			RW - 0x00					RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset

21	ANFF	When set to 1, frames not filtered in time and over the DMA RX FIFO level defined in THRESHOLD[4:0], are routed to the default RX FIFO Queue (defined by the ANMF_FQ[2:0] bit field). It is mandatory to have the default RX FIFO Queue defined in the ANMF_FQ[2:0] bit field, enabled and started (see RX_FQ_CTRL2 and RX_FQ_CTRL0 registers). This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0
20	ANMF	When set to 1, non matching frames are accepted, otherwise rejected. It is mandatory to have the default RX FIFO Queue defined in the ANMF_FQ[2:0] bit field, enabled and started (see RX_FQ_CTRL2 and RX_FQ_CTRL0 registers). This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0
18:16	ANMF_FQ	Define the default RX FIFO Queue number (from 0 to 7) when non matching frames are accepted (ANMF = 1) AND/OR when the threshold mechanism is active (ANFF = 1). This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x0
12:8	THRESHOLD	THRESHOLD defines the latest point in time to wait for the result of the RX filtering process., Once this limit is reached, the MH starts fetching an RX descriptor from S_MEM. THRESHOLD value is only used when greater than 0 and ANFF bit set to 1. See chapter "RX Filter" for an explanation how to configure THRESHOLD. When the RX filtering is not providing the result before the threshold of the RX DMA FIFO is reached, the RX message is sent to the default RX FIFO Queue mentioned in the ANMF_FQ[2:0] (only enabled when ANFF set to 1). When the level is over the threshold and the RX filtering result is already known, no action is taken. Threshold is given in number of word of 32bit. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00
7:0	NB_FE	Define the number of RX filter elements that are defined in the local memory. When set to 0, all RX messages are accepted and stored in the RX FIFO Queue number defined by ANMF_FQ[3:0]. This bit field register is only accessible in write mode if the MH is not busy, see BUSY flag in MH_STS register	RW - -	0x00

Register: TX_FQ_INT_STS

Description : TX FIFO Queue Interrupt Status register
Offset : 0x700
Absolute Address : 0x700
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								UNVALID							
								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								SENT							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
23:16	UNVALID	When TX FIFO Queue n loads a TX descriptor with VALID = 0, the bit UNVALID[n] will be set. Writing 1 to UNVALID[n] clears the bit.	RW - -	0x00
7:0	SENT	When SENT[n] = 1, a TX message was sent from the TX FIFO Queue n and writing a 1 clears the bit	RW - -	0x00

Register: RX_FQ_INT_STS

Description : RX FIFO Queue Interrupt Status register
Offset : 0x704
Absolute Address : 0x704
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								UNVALID							
								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								RECEIVED							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
23:16	UNVALID	When RX FIFO Queue n loads an RX descriptor with VALID=0, the bit UNVALID[n] will be set. Writing 1 to UNVALID[n] clears the bit.	RW - -	0x00
7:0	RECEIVED	When RECEIVED[n] = 1, an RX message was received in the RX FIFO Queue n, writing a 1 clears the bit	RW - -	0x00

Register: TX_PQ_INT_STS0

Description : TX Priority Queue Interrupt Status register 0
Offset : 0x708
Absolute Address : 0x708
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SENT															
RW - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SENT															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	SENT	When SENT[n] = 1 TX message was sent from the TX Priority Queue slot n, writing a 1 clears the bit	RW - -	0x0000 0000

Register: TX_PQ_INT_STS1

Description : TX Priority Queue Interrupt Status register 1
Offset : 0x70C
Absolute Address : 0x70C
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UNVALID															
RW - 0x0000 0000															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UNVALID															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	UNVALID	When UNVALID[n] = 1, an invalid RX descriptor is detected while running the TX Priority Queue slot n. Writing a 1 clears the bit. When set to 1, the TX Priority Queue slot n is on hold, waiting for the SW to react. As the TX message is fully defined in system memory before starting the relevant slot, there should not be any invalid TX descriptor interrupts	RW - -	0x0000 0000

Register: STATS_INT_STS

Description : Statistics Interrupt Status register
Offset : 0x710
Absolute Address : 0x710
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
												RX_UNSUCC	RX_SUCC	TX_UNSUCC	TX_SUCC
												RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
3	RX_UNSUCC	Counter of RX message received unsuccessfully has wrapped, writing a 1 clears the bit	RW - -	0x0
2	RX_SUCC	Counter of RX message received successfully has wrapped, writing a 1 clears the bit	RW - -	0x0

1	TX_UNSUCC	Counter of TX message transmitted unsuccessfully has wrapped, writing a 1 clears the bit	RW - -	0x0
0	TX_SUCC	Counter of TX message transmitted successfully has wrapped, writing a 1 clears the bit	RW - -	0x0

Register: ERR_INT_STS

Description : Error Interrupt Status register
Offset : 0x714
Absolute Address : 0x714
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
											DP_RX_SEQ_ERR	DP_TX_SEQ_ERR	DP_RX_ACK_DO_ERR	DP_RX_FIFO_DO_ERR	DP_TX_ACK_DO_ERR
											RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
4	DP_RX_SEQ_ERR	When set to 1, an RX sequence issue is detected, writing a 1 clears the bit	RW - -	0x0
3	DP_TX_SEQ_ERR	When set to 1, a TX sequence issue is detected, writing a 1 clears the bit	RW - -	0x0
2	DP_RX_ACK_DO_ERR	When set to 1, an RX acknowledge data overflow is detected, writing a 1 clears the bit	RW - -	0x0
1	DP_RX_FIFO_DO_ERR	When set to 1, an RX DMA FIFO overflow issue is detected, writing a 1 clears the bit	RW - -	0x0
0	DP_TX_ACK_DO_ERR	When set to 1, a TX acknowledge data overflow is detected, writing a 1 clears the bit	RW - -	0x0

Register: SFTY_INT_STS

Description : Safety Interrupt Status register
Offset : 0x718
Absolute Address : 0x718
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
														ACK_RX_PARITY_ERR	ACK_TX_PARITY_ERR
														RW - 0x0	RW - 0x0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MEM_SFTY_CE	MEM_SFTY_UE	RX_DESC_CRC_ERR	RX_DESC_REQ_ERR	TX_DESC_CRC_ERR	TX_DESC_REQ_ERR	AP_RX_PARITY_ERR	AP_TX_PARITY_ERR	DP_RX_PARITY_ERR	DP_TX_PARITY_ERR	MEM_AXI_RD_TO_ERR	MEM_AXI_WR_TO_ERR	DP_PRT_RX_TO_ERR	DP_PRT_TX_TO_ERR	DMA_AXI_RD_TO_ERR	DMA_AXI_WR_TO_ERR
RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
17	ACK_RX_PARITY_ERR	When set to 1, an acknowledge data parity issue is detected on the RX path	RW - -	0x0
16	ACK_TX_PARITY_ERR	When set to 1, an acknowledge data parity issue is detected on the TX path	RW - -	0x0
15	MEM_SFTY_CE	When set to 1, a correctable error is detected on the local memory interface	RW - -	0x0
14	MEM_SFTY_UE	When set to 1, an uncorrectable error is detected on the local memory interface	RW - -	0x0
13	RX_DESC_CRC_ERR	When set to 1, an RX descriptor has a wrong CRC, writing a 1 clears the bit	RW - -	0x0
12	RX_DESC_REQ_ERR	When set to 1, an RX descriptor fetched does not match the one expected, writing a 1 clears the bit	RW - -	0x0

11	TX_DESC_CRC_ERR	When set to 1, a TX descriptor has a wrong CRC, writing a 1 clears the bit	RW - -	0x0
10	TX_DESC_REQ_ERR	When set to 1, a TX descriptor fetched does not match the one expected, writing a 1 clears the bit	RW - -	0x0
9	AP_RX_PARITY_ERR	When set to 1, an RX address pointer parity issue is detected, writing a 1 clears the bit	RW - -	0x0
8	AP_TX_PARITY_ERR	When set to 1, a TX address pointer parity issue is detected, writing a 1 clears the bit	RW - -	0x0
7	DP_RX_PARITY_ERR	When set to 1, an RX data parity error is detected on datapath, writing a 1 clears the bit	RW - -	0x0
6	DP_TX_PARITY_ERR	When set to 1, a TX data parity error is detected on datapath, writing a 1 clears the bit	RW - -	0x0
5	MEM_AXI_RD_TO_ERR	When set to 1, an AXI read access timeout issue is detected on local memory interface, writing a 1 clears the bit	RW - -	0x0
4	MEM_AXI_WR_TO_ERR	When set to 1, an AXI write access timeout issue is detected on local memory interface, writing a 1 clears the bit	RW - -	0x0
3	DP_PRT_RX_TO_ERR	When set to 1, an RX_MSG timeout issue is detected, writing a 1 clears the bit	RW - -	0x0
2	DP_PRT_TX_TO_ERR	When set to 1, a TX_MSG timeout issue is detected, writing a 1 clears the bit	RW - -	0x0
1	DMA_AXI_RD_TO_ERR	When set to 1, an AXI read access timeout issue is detected on DMA interface, writing a 1 clears the bit	RW - -	0x0
0	DMA_AXI_WR_TO_ERR	When set to 1, an AXI write access timeout issue is detected on DMA interface, writing a 1 clears the bit	RW - -	0x0

Register: AXI_ERR_INFO

Description	: DMA Error Information
Offset	: 0x71C
Absolute Address	: 0x71C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								MEM_RESP		MEM_ID		DMA_RESP		DMA_ID	
								RO - 0x0		RO - 0x0		RO - 0x0		RO - 0x0	

Bit	Name	Description	SW Access HW Access Protection	Reset
7:6	MEM_RESP	On MEM_AXI interface. When set to 0b10, the AXI response is SLVERR. When set to 0b11, the response is DECERR. By default, set to 0b00 (OKAY)	RO - -	0x0
5:4	MEM_ID	On MEM_AXI interface. Define the AXI ID used when a write or read error response is detected. According to the value, the DMA channel can be identified and so it is possible to define what's the effect of such issue.	RO - -	0x0
3:2	DMA_RESP	On DMA_AXI interface. When set to 0b10, the AXI response is SLVERR. When set to 0b11, the response is DECERR. By default, set to 0b00 (OKAY)	RO - -	0x0
1:0	DMA_ID	On DMA_AXI interface. Define the AXI ID used when a write or read error response is detected. According to the value, the DMA channel can be identified and so it is possible to define what's the effect of such issue.	RO - -	0x0

Register: DESC_ERR_INFO0

Description	: Descriptor Error Information 0 If the DESC_ERR_INFO0.ADD[31:16] = 0 and DESC_ERR_INFO1.CRC[8:0], DESC_ERR_INFO1.RX_TX and DESC_ERR_INFO1.RC[4:0] are all equal to 0, the faulty descriptor is a TX descriptor fetched from L_MEM
Offset	: 0x720
Absolute Address	: 0x720
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADD															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADD															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	ADD	Descriptor address being used when the error is detected	RO - -	0x0000 0000

Register: DESC_ERR_INFO1

Description	: Descriptor Error Information 1 When the DESC_ERR_INFO1.CRC[8:0], DESC_ERR_INFO1.RX_TX and DESC_ERR_INFO1.RC[4:0] are all equal to 0, the faulty descriptor is a TX descriptor fetched from L_MEM only if the DESC_ERR_INFO0.ADD[31:16] = 0
Offset	: 0x724
Absolute Address	: 0x724
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
							CRC								
							RO - 0x000								
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
RX_TX		RC					PQ	IN			FQN_PQSN				
RO - 0x0		RO - 0x00					RO - 0x0	RO - 0x0			RO - 0x00				

Bit	Name	Description	SW Access HW Access Protection	Reset
24:16	CRC	CRC value defined in the RX or TX descriptor logged in	RO - -	0x000
15	RX_TX	RX descriptor has an issue (RX_TX set to 1), otherwise the same for a TX descriptor	RO - -	0x0
13:9	RC	Provide the information regarding the Rolling Counter defined in RX or TX descriptor impacted	RO - -	0x00
8	PQ	Identify which TX queue is impacted, either the TX Priority Queue (PQ set to 1) or the TX FIFO Queues	RO - -	0x0
7:5	IN	Provide the instance number defined in RX or TX descriptor logged in	RO - -	0x0

4:0	FQN_PQSN	Provide the information regarding the RX/TX FIFO Queue number or the TX Priority Queue slot having an issue	RO - -	0x00

Register: TX_FILTER_ERR_INFO

Description	: TX Filter Error Information
Offset	: 0x728
Absolute Address	: 0x728
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
										FQN_PQS					FQ
										RO - 0x00					RO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
5:1	FQN_PQS	Provide the information of the TX FIFO Queue number or TX Priority Queue slot number which has triggered the TX_FILTER_ERR interrupt.	RO - -	0x00
0	FQ	When set to 1, one of the TX FIFO Queues has triggered the TX_FILTER_ERR interrupt, otherwise it is a TX Priority Queue slot	RO - -	0x0

Register: DEBUG_TEST_CTRL

Description	: Debug Control register This register is only accessible in write mode if the Test Mode Key sequence has been performed prior to write. The register is accessible in write access in privileged mode only. This register is protected by a register bank CRC defined in CRC_REG register.
Offset	: 0x800
Absolute Address	: 0x800
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
					HDP_SEL									HDP_EN	TEST_IRQ_EN
					RW - 0x0									RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
10:8	HDP_SEL	Define the set of signals to be monitored on the HDP[15:0] bus signal interface. This bit field register is only accessible in write mode if the Test Mode Key sequence has been performed prior to write.	RW - -	0x0
1	HDP_EN	When writing 1, enable the hardware debug port to monitor MH internal signals. This bit field register is only accessible in write mode if the Test Mode Key sequence has been performed prior to write.	RW - -	0x0
0	TEST_IRQ_EN	When writing 1, enable the control of the interrupt lines using the INT_TEST0 and INT_TEST1 registers. This bit field register is only accessible in write mode if the Test Mode Key sequence has been performed prior to write.	RW - -	0x0

Register: INT_TEST0

Description	: Interrupt Test register 0 This register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set
Offset	: 0x804
Absolute Address	: 0x804
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								RX_FQ_IRQ							
								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								TX_FQ_IRQ							
								RW - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
23:16	RX_FQ_IRQ	When writing 1 to RX_FQ_IRQ[n], triggers the interrupt line rx_fq_irq[n], those bits are auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x00
7:0	TX_FQ_IRQ	When writing 1 to TX_FQ_IRQ[n], triggers the interrupt line tx_fq_irq[n], those bits are auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x00

Register: INT_TEST1

Description	: Interrupt Test register 1 This register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set
Offset	: 0x808
Absolute Address	: 0x808
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
											TX_PQ_IRQ	STATS_IRQ	STOP_IRQ	RX_FILTER_IRQ	TX_FILTER_IRQ
											RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
TX_ABORT_IRQ	RX_ABORT_IRQ	RX_FILTER_ERR	MEM_TO_ERR	MEM_SFTY_ERR	REG_CRC_ERR	DESC_ERR	AP_PARITY_ERR	DP_PARITY_ERR	DP_SEQ_ERR	DP_DO_ERR	DP_TO_ERR	DMA_CH_ERR	DMA_TO_ERR	RESP_ERR	
RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	

Bit	Name	Description	SW Access HW Access Protection	Reset

20	TX_PQ_IRQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
19	STATS_IRQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
18	STOP_IRQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
17	RX_FILTER_IRQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
16	TX_FILTER_IRQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
15	TX_ABORT_IRQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
14	RX_ABORT_IRQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
13	RX_FILTER_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
12	MEM_TO_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
11	MEM_SFTY_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0

10	REG_CRC_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
9	DESC_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
8	AP_PARITY_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
7	DP_PARITY_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
6	DP_SEQ_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
5	DP_DO_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
4	DP_TO_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
3	DMA_CH_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
2	DMA_TO_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0
1:0	RESP_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared. This bit field register is only accessible in write mode if the TEST_IRQ_EN bit in DEBUG_TEST_CTRL is set	RW - -	0x0

Register: TX_SCAN_FC

Description	: TX-SCAN first candidates register This register gives the 4 best candidates evaluated by the TX-Scan. This register gives the first and second highest priority TX descriptor after a TX-Scan. It provides also the third and fourth candidates during a TX-Scan, considering the first and second candidates as already defined by a previous TX-Scan run.
Offset	: 0x810
Absolute Address	: 0x810
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
		FQN_PQSN3					FQ_PQ3			FQN_PQSN2					FQ_PQ2
		RO - 0x00					RO - 0x0			RO - 0x00					RO - 0x0
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
		FQN_PQSN1					FQ_PQ1			FQN_PQSN0					FQ_PQ0
		RO - 0x00					RO - 0x0			RO - 0x00					RO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
29:25	FQN_PQSN3	The fourth candidate is coming from either the TX FIFO Queue number N (defined by FQN in TX descriptor) or the TX Priority Queue Slot number M (defined by the PQSN in TX descriptor). The meaning of this bit field depends on the PQ3.	RO - -	0x00
24	FQ_PQ3	The fourth candidate evaluated by TX-Scan is either a TX Priority Queue (when set to 1) or a TX FIFO Queue (when set to 0).	RO - -	0x0
21:17	FQN_PQSN2	The third candidate is coming from either the TX FIFO Queue number N (defined by FQN in TX descriptor) or the TX Priority Queue Slot number M (defined by the PQSN in TX descriptor). The meaning of this bit field depends on the PQ2.	RO - -	0x00
16	FQ_PQ2	The third candidate evaluated by TX-Scan is either a TX Priority Queue (when set to 1) or a TX FIFO Queue (when set to 0).	RO - -	0x0

13:9	FQN_PQSN1	The second candidate is coming from either the TX FIFO Queue number N (defined by FQN in TX descriptor) or the TX Priority Queue Slot number M (defined by the PQSN in TX descriptor). The meaning of this bit field depends on the PQ0. This bit field is identical to the TX_SCAN_BC.SH_FQN_PQSN bit register	RO - -	0x00
8	FQ_PQ1	The second candidate evaluated by TX-Scan is either a TX Priority Queue (when set to 1) or a TX FIFO Queue (when set to 0). This bit field is identical to TX_SCAN_BC.SH_PQ bit register	RO - -	0x0
5:1	FQN_PQSN0	The first candidate is coming from either the TX FIFO Queue number N (defined by FQN in TX descriptor) or the TX Priority Queue Slot number M (defined by the PQSN in TX descriptor). The meaning of this bit field depends on the PQ0. This bit field is identical to TX_SCAN_BC.FH_FQN_PQSN bit register	RO - -	0x00
0	FQ_PQ0	The first candidate evaluated by TX-Scan is either a TX Priority Queue (when set to 1) or a TX FIFO Queue (when set to 0). This bit field is identical to TX_SCAN_BC.FH_PQ bit register	RO - -	0x0

Register: TX_SCAN_BC

Description	: TX-SCAN best candidates register This register gives the first and second highest priority TX descriptor after a TX-Scan
Offset	: 0x814
Absolute Address	: 0x814
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SH_OFFSET										SH_FQN_PQSN					SH_PQ
RO - 0x000										RO - 0x00					RO - 0x0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FH_OFFSET										FH_FQN_PQSN					FH_PQ
RO - 0x000										RO - 0x00					RO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
31:22	SH_OFFSET	Second highest priority candidate offset in multiple of 32bytes (TX descriptor size). This register is relevant only for the TX FIFO Queue. It provides the index of the TX descriptor in the TX FIFO Queue which is about to be sent on the CAN bus. When SH_PQ = 1 it is set to 0.	RO - -	0x000
21:17	SH_FQN_PQSN	Second highest priority candidate coming from either the TX FIFO Queue number N (defined by FQN in TX descriptor) or the TX Priority Queue Slot number M (defined by the PQSN in TX descriptor). The meaning of this bit field depends on the SH_PQ.	RO - -	0x00
16	SH_PQ	Second highest priority candidate evaluated by TX-Scan. It is either a TX Priority Queue (when set to 1) or a TX FIFO Queue (when set to 0).	RO - -	0x0
15:6	FH_OFFSET	First highest priority candidate offset in multiple of 32bytes (TX descriptor size). This register is relevant only for the TX FIFO Queue. It provides the index of the TX descriptor in the TX FIFO Queue which is in use on the CAN bus. When FH_PQ = 1 it is set to 0.	RO - -	0x000
5:1	FH_FQN_PQSN	First highest priority candidate coming from either the TX FIFO Queue number N (defined by FQN in TX descriptor) or the TX Priority Queue Slot number M (defined by the PQSN in TX descriptor). The meaning of this bit field depends on the FH_PQ.	RO - -	0x00
0	FH_PQ	First highest priority candidate evaluated by TX-Scan. It is either a TX Priority Queue (when set to 1) or a TX FIFO Queue (when set to 0).	RO - -	0x0

Register: TX_FQ_DESC_VALID

Description	: Valid TX FIFO Queue descriptors in local memory
Offset	: 0x818
Absolute Address	: 0x818
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
								DESC_NC_VALID							
								RO - 0x00							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
								DESC_CN_VALID							
								RO - 0x00							

Bit	Name	Description	SW Access HW Access Protection	Reset
23:16	DESC_NC_VALID	When DESC_NC_VALID[n] = 1, the next/current TX descriptor for the TX FIFO Queue is available in L_MEM	RO - -	0x00
7:0	DESC_CN_VALID	When DESC_CN_VALID[n] = 1, the current/next TX descriptor for the TX FIFO Queue n is available in L_MEM	RO - -	0x00

Register: TX_PQ_DESC_VALID

Description : Valid TX Priority Queue descriptors in local memory
Offset : 0x81C
Absolute Address : 0x81C
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DESC_VALID															
RO - 0x0000 0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DESC_VALID															
RO - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	DESC_VALID	When DESC_VALID[n] = 1, the TX descriptor assigned to the slot n in local memory is valid	RO - -	0x0000 0000

Register: CRC_CTRL

Description : CRC Control register
Offset : 0x880
Absolute Address : 0x880
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
															STA RT
															WO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
0	START	Writing a 1 to this bit triggers the HW CRC check of registers. This action can be done any time for a sanity check	WO - -	0x0

Register: CRC_REG

Description	: CRC register
Offset	: 0x884
Absolute Address	: 0x884
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VAL															
RW - 0x0000 0000															
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
VAL															
RW - 0x0000 0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	VAL	CRC value of all the registers protected by CRC. Once done, a write to the START bit in the CRC_CTRL register must be done	RW - -	0x0000 0000

Block: xcan_prt_inst_0__reg_axi__900__status

Description : Status information of the PRT

Register: ENDN

Description : Endianness Test Register
Offset : 0x0
Absolute Address : 0x900
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ETV															
RO - 0x8765 4321															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETV															
RO - 0x8765 4321															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:0	ETV	The purpose of this register is to identify the beginning of the PRT address map in a memory dump and to check the proper endianness data byte mapping when the data word is routed through different busses.	RO - -	0x8765 4321

Register: PREL

Description : PRT Release Identification Register.
Offset : 0x4
Absolute Address : 0x904
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REL				STEP				SUBSTEP				YEAR			
RO - 0x0				RO - 0x5				RO - 0x4				RO - 0x0			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

MON	DAY
RO - 0x00	RO - 0x00

Bit	Name	Description	SW Access HW Access Protection	Reset
31:28	REL	Release value, used to identify the main release of the XCAN_PRT.	RO - -	0x0
27:24	STEP	Step value according to Release	RO - -	0x5
23:20	SUBSTEP	Sub-Step value according to Step	RO - -	0x4
19:16	YEAR	Define the year of the release using a binary coded decimal representation (0 being 2020 and so forth…). This reset value is defined by the generic parameter DESIGN_TIME_STAMP_G[19:16]. If the generic parameter DESIGN_TIME_STAMP_G is not set, the default value is the one defined here	RO - -	0x0
15:8	MON	Define the month of the release using a binary coded decimal representation (1 being January and so forth). This reset value is defined by the generic parameter DESIGN_TIME_STAMP_G[15:8]. If the generic parameter DESIGN_TIME_STAMP_G is not set, the default value is the one defined here	RO - -	0x00
7:0	DAY	Define the day of the release using a binary coded decimal representation (1 being the first day of the month and so forth). This reset value is defined by the generic parameter DESIGN_TIME_STAMP_G[7:0]. If the generic parameter DESIGN_TIME_STAMP_G is not set, the default value is the one defined here	RO - -	0x00

Register: STAT

Description	: PRT Status Register
Offset	: 0x8
Absolute Address	: 0x908
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

TEC								RP	REC						
RO - 0x00								RO - 0x0	RO - 0x00						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDCV								BO	EP	FIMA	CLKA	STP	INT	ACT	
RO - 0x00								RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x1	RO - 0x0	RO - 0x0	RO - 0x0	

Bit	Name	Description	SW Access HW Access Protection	Reset
31:24	TEC	The CAN protocol's Transmit Error Counter. A software reset does not change the value in this register. When the Bus-Off recovery sequence is finished, the error counter TEC is cleared. When the increment TEC+8 would result in a value > 255 (carry-flag), the TEC is kept unchanged, but BO is set. The Transmit Error Counter is decremented by one each time a CAN message has been successfully transmitted, but it is not decremented below the value 0.	RO - -	0x00
23	RP	The Passive flag of the CAN protocol's Receive Error Counter. This flag is set on an error condition that would have caused an increment of the Receive Error Counter to a value beyond its 7 bit range.	RO - -	0x0
22:16	REC	The CAN protocol's Receive Error Counter. A software reset does not change the value in this register. When the Bus-Off recovery sequence is finished, the error counter REC is cleared. The REC is a 7-bit-counter, together with the Error-Passive flag EP. When the increment REC+1 or REC+8 would result in a value > 127 (carry-flag), the REC is kept unchanged, but EP is set. When EP is set but REC is below 127 and further errors are detected with an REC+1 condition, the REC will be incremented until it reaches 127. At the reception of a valid message, the REC-1 decrements the actual value of the REC by one AND clears the Error-Passive flag EP.	RO - -	0x00
15:8	TDCV	Transmitter Delay Compensation's delay value. A software reset clears the TDV bit field to 0x00. This register shows the sum of the measured delay plus the configured offset, giving the position of the secondary sample point. It is updated for each frame transmission that includes a data phase.	RO - -	0x00

7	BO	This node is in Bus-Off state. This flag is set on an error condition that would have caused an increment of the Transmit Error Counter to a value beyond its 8 bit range. When the PRT enters Bus-Off state, BO is set to 1 and CAN protocol operation is stopped. When the Bus-Off recovery sequence is finished, BO is cleared.	RO - -	0x0
6	EP	This node is in Error-Passive state. When both error counters drop below 127, or when the Bus-Off recovery sequence is finished, the EP bit is cleared.	RO - -	0x0
5	FIMA	Fault Injection Module Activated, see Safety Measures chapter	RO - -	0x0
4	CLKA	The actual value of the CLOCK_ACTIVE input signal, see Starting and Stopping the Module chapter. As the clock must be active when a reset is performed, the default value should be 1.	RO - -	0x1
3	STP	Waiting for end of actual message after STOP command, see Starting and Stopping The Module chapter	RO - -	0x0
2	INT	This node is integrating into CAN bus traffic	RO - -	0x0

1:0	ACT	The current activity of this node: 0b00: inactive state 0b01: Idle 0b10: Receiver 0b11: Transmitter When the CAN protocol operation is stopped, ACT changes to 0b00 and INT changes to 0. When the CAN protocol operation is started, INT is set to 1, but ACT remains at 0b00 until the CAN protocol's bus idle detection condition is met, then it changes to 0b01 and INT changes to 0. When the CAN protocol operation is started while BO is set, the PRT remains in integrating state (INT=1 and ACT=0b00) until the Bus-Off recovery sequence is finished, then it changes to 0b01 and INT changes to 0. When PRT detects a protocol exception event (see [1], chapter 10.9.5), ACT changes to 0b00 and INT changes to 1 until the CAN protocol's bus idle detection condition is met, then ACT changes to 0b01 and INT changes to 0. ACT changes from 0b01 to 0b10 when the PRT has received a Start-of-Frame from the CAN bus. ACT changes from 0b01 to 0b11 when the PRT has sent a Start-of-Frame to the CAN bus. ACT changes from 0b11 to 0b10 when the PRT loses arbitration during a transmission. ACT changes from 0b10 to 0b01 or from 0b11 to 0b01when the PRT detects the second bit of intermission (see [1], chapter 10.4.6.2) to be recessive.	RO - -	0x0

Block: xcan_prt_inst_0__reg_axi__900__event

Description : Event information of the PRT

Register: EVNT

Description : Event Status Flags Register
The EVNT Register contains event status flags. The flags are set by the PRT when specific events occur. A software reset clears all flags. A host write access to this register, writing a 1 to a specific flag, clears that flag. When a host write access occurs concurrently with a set condition for a flag, the flag is set.

Offset : 0x0

Absolute Address : 0x920

Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		ABO	IFR	USO	DU	PXE	TXF	RXF	DO	STE	FRE	AKE	B1E	B0E	CRE
		W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0	W1C - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
13	ABO	TX_MSG sequence stopped by TX_MSG_WUSER code ABORT	W1C - -	0x0
12	IFR	Invalid Frame Format requested in TX_MSG	W1C - -	0x0
11	USO	Unexpected Start of Sequence during TX_MSG sequence detected	W1C - -	0x0
10	DU	Underrun condition in TX_MSG sequence detected	W1C - -	0x0
9	PXE	Protocol Exception Event occurred	W1C - -	0x0

8	TXF	Frame transmitted	W1C - -	0x0
7	RXF	Frame received	W1C - -	0x0
6	DO	Overflow condition in RX_MSG sequence detected	W1C - -	0x0
5	STE	Stuff Error	W1C - -	0x0
4	FRE	Form Error or the condition of CAN error counting rule f)	W1C - -	0x0
3	AKE	Acknowledge Error	W1C - -	0x0
2	B1E	Bit1Error: During the transmission of a message (with the exception of the arbitration field), the PRT wanted to send a recessive bit (logical value 1), but the monitored CAN bus value was dominant.	W1C - -	0x0
1	B0E	Bit0 Error: The PRT wanted to send a dominant bit (logical value 0), but the monitored CAN bus value was recessive. During Bus-Off recovery, B0E is also set each time a sequence of 11 recessive bits has been monitored, enabling the CPU to readily check whether the CAN bus is stuck at dominant or continuously disturbed, and to monitor the proceeding of the Bus-Off recovery sequence.	W1C - -	0x0
0	CRE	CRC Error	W1C - -	0x0

Block: xcan_prt_inst_0__reg_axi__900__control

Description	: Control of the PRT during runtime

Register: LOCK

Description	: Unlock Sequence Register Writing a sequence of specific data words enables the activation of control commands in the registers CTRL and FIMC. Reading this register always gives the value 0x00000000.
Offset	: 0x0
Absolute Address	: 0x940
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TMK															
WO - 0x0000															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ULK															
WO - 0x0000															

Bit	Name	Description	SW Access HW Access Protection	Reset
31:16	TMK	Test Mode Key	WO - -	0x0000
15:0	ULK	Unlock Key	WO - -	0x0000

Register: CTRL

Description	: Control Register Writing to this register controls the CAN protocol operation. Reading this register gives the value 0x00000000. When writing to this register, only one of the four bits TEST, SRES , STRT, or STOP may be written to 1, otherwise the write access takes no effect. The bit IMMD may be written to 1 together with the bit STOP, but not together with one of the other bits.
Offset	: 0x4

Absolute Address : 0x944
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
			TEST				SRES				STRT			IMMD	STOP
			WO - 0x0				WO - 0x0				WO - 0x0			WO - 0x0	WO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
12	TEST	Enable Test Mode. The Test Mode Key must be used prior to write to this bit field.	WO - -	0x0
8	SRES	Software Reset. When the CAN protocol operation is stopped, the software reset of all state machines of the PRT (excluding the error-counters and the error-states) is triggered by writing 1 to CTRL.SRES. No unlocking sequence is required. A software reset will not be executed while the CAN protocol operation is started.	WO - -	0x0
4	STRT	Start CAN protocol operation.	WO - -	0x0
1	IMMD	Stop CAN protocol operation immediately. The Unlock Key must be used prior to write to this bit . This bit is only effective when being set together with the bit STOP.	WO - -	0x0
0	STOP	Stop CAN protocol operation. The Unlock Key must be used prior to write to this bit field. When not set together with bit IMMD the PRT waits for an ongoing CAN message to finish before stopping CAN protocol operation.	WO - -	0x0

Register: FIMC

Description : Fault Injection Module Control Register
Writing the fault injection position number requires the application of the test mode key sequence before writing to FIMC. This register must be accessed in privileged mode when supported.
Offset : 0x8
Absolute Address : 0x948

Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
	FIP														
	RW - 0x0000														

Bit	Name	Description	SW Access HW Access Protection	Reset
14:0	FIP	Fault Injection Position. Writing to FIMC while MODE.FIME is set activates the Fault Injection Module FIM (see Safety Measures chapter). While the FIM is activated, the value of FIMC.FIP is protected from further write accesses until the FIM is de-activated again.	RW - -	0x0000

Register: TEST

Description : Hardware Test Functions Register
This register is writable after the hardware test mode functions are enabled by writing the test mode key sequence to LOCK and CTRL registers. While the hardware test mode functions are not enabled, this register is read-only. This register must be accessed in privileged mode when supported. The hardware test mode functions are disabled and cleared by the software reset of the PRT.

Offset : 0xC

Absolute Address : 0x94C

Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				BUS_OFF	BUS_ON	E_PASSIVE	E_ACTIVE	BUS_ERR	RX_EVT	TX_EVT	IFF_RQ	RX_DO	TX_DU	USOS	ABORTED
				WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
HWT										TXC		RXD			LBCK
RO - 0x0										RW - 0x0		RO - 0x1			RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
27	BUS_OFF	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
26	BUS_ON	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
25	E_PASSIVE	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
24	E_ACTIVE	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
23	BUS_ERR	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
22	RX_EVT	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
21	TX_EVT	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
20	IFF_RQ	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
19	RX_DO	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
18	TX_DU	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
17	USOS	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
16	ABORTED	Writing a 1 to the bit field triggers the related interrupt line, this bit is auto-cleared	WO - -	0x0
15	HWT	This status flag HWT shows whether the hardware test mode functions are enabled, set to 1 means enable.	RO - -	0x0
5:4	TXC	Control the bit value driven at CAN_TX 0b00: Normal function of CAN TX 0b01: Normal function of CAN TX. CAN RX is ignored (for message look back mode) 0b10: CAN TX output set to 0 0b11: CAN TX output set to 1	RW - -	0x0

3	RXD	Bit value seen at CAN_RX. The CAN_RX input (output signal of the transceiver) is always readable through this bit.	RO - -	0x1
0	LBCK	Enable the Message Loop-Back mode, see chapter Trace and Debug.	RW - -	0x0

Block: xcan_prt_inst_0__reg_axi__900__configuration

Description : Configuration of the PRT before runtime

Register: MODE

Description : Operating Mode Register
Configuration register that is writable while the CAN communication is stopped and that is read-only after the CAN communication is started. This register defines separate operating mode options. The four configuration bits FDOE, XLOE, EFDI, and XLTR, are interrelated according to table Frame Formats defined in Operating Mode chapter.

Offset : 0x0

Absolute Address : 0x960

Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				FIME	EFDI	XLTR	SFS	RSTR	MON	TXP	EFBI	PXHD	TDCE	XLOE	FDOE
				RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
11	FIME	Fault Injection Module Enable, see Safety Measures chapter	RW - -	0x0
10	EFDI	Error Flag Disable, 1 means Error Signalling is disabled as defined in [2] and the error counters REC and TEC are not incremented. When this bit is set, only CAN XL frames are transmitted and received dominant FDF or XLF bits are treated as form errors.	RW - -	0x0
9	XLTR	XL Transceiver Connected	RW - -	0x0

8	SFS	Time stamp position: Start of Frame Stamping 1: Timestamps captured at the start of a frame 0: Timestamps captured at the end of a frame	RW - -	0x0
7	RSTR	Restricted Mode Enabled as defined in [1]	RW - -	0x0
6	MON	Monitoring Mode Enabled as defined in [1]	RW - -	0x0
5	TXP	Transmit Pause. If this bit is set, the PRT pauses for two CAN bit times before starting the next transmission after itself has successfully transmitted a frame	RW - -	0x0
4	EFBI	Edge Filtering during Bus Integration. If this bit is set, the PRT requires two consecutive dominant to to detect an edge causing the reset of the bit counter for the detection of the idle condition.	RW - -	0x0
3	PXHD	Protocol Exception Handling Disabled	RW - -	0x0
2	TDCE	Transmitter Delay Compensation Enabled as defined in [1]	RW - -	0x0
1	XLOE	XL Frame Format enabled. When set to 0, node behaves according to ISO11898-1:2015, no arbitration during FDF bit. When set to 1, node behaves according to CiA610-1, arbitration during FDF bit and XLF bit. This bit cannot be set to 1 when one of the static inputs ONLY_CC or ONLY_CC_FD is set. Setting XLOE without setting FDOE is an invalid configuration.	RW - -	0x0
0	FDOE	FD Frame Format enabled. When set to 1, node is FD enabled according to ISO11898-1:2015. When set to 0, node is FD tolerant according to ISO11898-1:2015 (only Classical CAN frames used). This bit cannot be set to 1 when the static input ONLY_CC is set.	RW - -	0x0

Register: NBTP

Description	: Arbitration Phase Nominal Bit Timing Register Configuration register that is writable while the CAN communication is stopped and that is read-only after the CAN communication is started. This register defines the Nominal Bit Timing as defined in [1].
Offset	: 0x4
Absolute Address	: 0x964
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
		BRP						NTSEG1							
		RW - 0x00						RW - 0x000							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	NTSEG2								NSJW						
	RW - 0x00								RW - 0x00						

Bit	Name	Description	SW Access HW Access Protection	Reset
29:25	BRP	Bit Rate Prescaler. Valid values for the Bit Rate Prescaler BRP are 0x00-0x1F. This value defines the length of the Time Quantum TQ for all three bit time configurations. The actual interpretation of this value is that the TQ is (BRP + 1) CLK periods long	RW - -	0x00
24:16	NTSEG1	Nominal Prop_Seg and Phase_Seg1. Valid values for NTSEG1 are 0x01-0x1FF. This value defines the sum of Prop_Seg(N) and Phase_Seg1(N). The actual interpretation of this value is that these segments together are (NTSEG1 + 1) TQ long.	RW - -	0x000
14:8	NTSEG2	Nominal Phase_Seg2. Valid values for NTSEG2 are 0x01-0x7F. This value defines the length of Phase_Seg2(N). The actual interpretation of this value is that the phase buffer segment 2 is (NTSEG2 + 1) TQ long.	RW - -	0x00
6:0	NSJW	Nominal SJW. Valid values for the Nominal Synchronization Jump Width NSJW are 0x00-0x7F. The actual interpretation of this value is that the Nominal Synchronization Jump Width is (NSJW + 1) TQ long.	RW - -	0x00

Register: DBTP

Description	: CAN FD Data Phase Bit Timing Register Configuration register that is writable while the CAN communication is stopped and that is read-only after the CAN communication is started. This register defines the FD Data Phase Bit Timing as defined in [1].
Offset	: 0x8
Absolute Address	: 0x968
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DTDCO								DTSEG1							
RW - 0x00								RW - 0x00							
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
	DTSEG2								DSJW						
	RW - 0x00								RW - 0x00						

Bit	Name	Description	SW Access HW Access Protection	Reset
31:24	DTDCO	Transmitter Delay Compensation Offset for FD frames. Valid values for the FD Transmitter Delay Compensation Offset DTDCO is 0x00-0xFF. This configuration defines the distance between the measured delay from CAN_TX to CAN_RX and the secondary sample point SSP, measured in CLK periods. This value is used when transmitting a CAN FD frame	RW - -	0x00
23:16	DTSEG1	FD data phase Prop_Seg and Phase_Seg1. Valid values for DTSEG1 are 0x00-0xFF. This value defines the sum of Prop_Seg(D) and Phase_Seg1(D). The actual interpretation of this value is that these segments together are (DTSEG1 + 1) TQ long	RW - -	0x00
14:8	DTSEG2	FD data phase Phase_Seg2. Valid values for DTSEG2 are 0x01-0x7F. This value defines the length of Phase_Seg2(D). The actual interpretation of this value is that the phase buffer segment 2 is (DTSEG2 + 1) TQ long.	RW - -	0x00
6:0	DSJW	FD data phase SJW. Valid values for the FD data phase Synchronization Jump Width DSJW are 0x00-0x7F. The actual interpretation of this value is that the FD data phase Synchronization Jump Width is (DSJW + 1) TQ long.	RW - -	0x00

Register: XBTP

Description	: CAN XL Data Phase Bit Timing Register Configuration register that is writable while the CAN communication is stopped and is read-only after the CAN communication is started. This register defines the XL Data Phase Bit Timing as defined in [2].
Offset	: 0xC
Absolute Address	: 0x96C
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
XTDCO								XTSEG1							
RW - 0x00								RW - 0x00							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	XTSEG2								XSJW						
	RW - 0x00								RW - 0x00						

Bit	Name	Description	SW Access HW Access Protection	Reset
31:24	XTDCO	Transmitter Delay Compensation Offset for XL frames. Valid values for the XL Transmitter Delay Compensation Offset XTDCO is 0x00-0xFF. This configuration defines the distance between the measured delay from CAN_TX to CAN_RX and the secondary sample point SSP, measured in CLK periods. This value is used when transmitting a CAN XL frame.	RW - -	0x00
23:16	XTSEG1	XL data phase Prop_Seg and Phase_Seg1. Valid values for XTSEG1 are 0x00-0xFF. This value defines the sum of Prop_Seg(X) and Phase_Seg1(X). The actual interpretation of this value is that these segments together are (XTSEG1 + 1) TQ long	RW - -	0x00
14:8	XTSEG2	XL data phase Phase_Seg2. Valid values for XTSEG2 are 0x01-0x7F. This value defines the length of Phase_Seg2(X). The actual interpretation of this value is that the phase buffer segment 2 is (XTSEG2 + 1) TQ long	RW - -	0x00
6:0	XSJW	XL data phase SJW. Valid values for the XL data phase Synchronization Jump Width XSJW are 0x00-0x7F. The actual interpretation of this value is that the XL data phase Synchronization Jump Width is (XSJW + 1) TQ long	RW - -	0x00

Register: PCFG

Description	: PWME Configuration Register Configuration register that is writable while the CAN communication is stopped and is read-only after the CAN communication is started. This register defines the parameters needed for the PWM coding (as described in [2]) in the PWME module for CAN XL transceivers with switchable operating modes
Offset	: 0x10
Absolute Address	: 0x970
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
										PWMO					
										RW - 0x00					
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
		PWML								PWMS					
		RW - 0x00								RW - 0x00					

Bit	Name	Description	SW Access HW Access Protection	Reset
21:16	PWMO	PWM Offset	RW - -	0x00
13:8	PWML	PWM phase Long	RW - -	0x00
5:0	PWMS	PWM phase Short	RW - -	0x00

Block: xcand_top_irc_inst_0__irc_axi__a00__Event

Description	: This address block provides registers to capture events of the MH and the PRT. The events are organized in three categories (one register for each category): Functional relevant, functional error relevant and safety relevant.

Register: FUNC_RAW

Description	: Functional raw event status register. This register provides information about the occurrence of functional relevant events inside the MH and the PRT. A flag is set when the related event is detected, independent of FUNC_ENA. The flags remain set until the Host CPU clears them by writing a 1 to the corresponding bit position at register FUNC_CLR.
Offset	: 0x0
Absolute Address	: 0xA00
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				PRT_RX_EVT	PRT_TX_EVT	PRT_BUS_ON	PRT_E_ACTIVE		MH_STATS_IRQ	MH_RX_ABORT_IRQ	MH_TX_ABORT_IRQ	MH_TX_FILTER_IRQ	MH_RX_FILTER_IRQ	MH_STOP_IRQ	MH_TX_PQ_IRQ
				RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0		RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
MH_RX_FQ7_IRQ	MH_RX_FQ6_IRQ	MH_RX_FQ5_IRQ	MH_RX_FQ4_IRQ	MH_RX_FQ3_IRQ	MH_RX_FQ2_IRQ	MH_RX_FQ1_IRQ	MH_RX_FQ0_IRQ	MH_TX_FQ7_IRQ	MH_TX_FQ6_IRQ	MH_TX_FQ5_IRQ	MH_TX_FQ4_IRQ	MH_TX_FQ3_IRQ	MH_TX_FQ2_IRQ	MH_TX_FQ1_IRQ	MH_TX_FQ0_IRQ
RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
27	PRT_RX_EVT	PRT received a valid CAN message	RO - -	0x0

26	PRT_TX_EVT	PRT transmitted a valid CAN message	RO - -	0x0
25	PRT_BUS_ON	PRT started CAN communication, after start or end of BusOff	RO - -	0x0
24	PRT_E_ACTIVE	PRT switched from Error-Passive to Error-Active state.	RO - -	0x0
22	MH_STATS_IRQ	One of the RX/TX counters have reached the threshold.	RO - -	0x0
21	MH_RX_ABORT_IRQ	This interrupt line is triggered when the MH needs to abort a RX message being received from PRT.	RO - -	0x0
20	MH_TX_ABORT_IRQ	This interrupt line is triggered when the MH needs to abort a TX message being sent to the PRT.	RO - -	0x0
19	MH_TX_FILTER_IRQ	The interrupt is triggered when the TX filter is enabled, and a TX message is rejected.	RO - -	0x0
18	MH_RX_FILTER_IRQ	In order to track RX filtering results, an interrupt can be triggered when the comparison between a RX message header and a defined filter is successful.	RO - -	0x0
17	MH_STOP_IRQ	The interrupt is triggered when the PRT is stopped. The MH finishes its task and switches to idle mode.	RO - -	0x0
16	MH_TX_PQ_IRQ	Interrupt of TX Priority Queue. Any TX message sent from the TX Priority Queue can be configured to trigger this interrupt. The SW would then need to look at the MH register TX_PQ_INT_STS to identify which slot has generated the interrupt and for which reason.	RO - -	0x0
15	MH_RX_FQ7_IRQ	MH interrupt of the RX FIFO Queue 7. Refer to the description of the MH_RX_FQ0_IRQ	RO - -	0x0
14	MH_RX_FQ6_IRQ	MH interrupt of the RX FIFO Queue 6. Refer to the description of the MH_RX_FQ0_IRQ	RO - -	0x0
13	MH_RX_FQ5_IRQ	MH interrupt of the RX FIFO Queue 5. Refer to the description of the MH_RX_FQ0_IRQ	RO - -	0x0
12	MH_RX_FQ4_IRQ	MH interrupt of the RX FIFO Queue 4. Refer to the description of the MH_RX_FQ0_IRQ	RO - -	0x0
11	MH_RX_FQ3_IRQ	MH interrupt of the RX FIFO Queue 3. Refer to the description of the MH_RX_FQ0_IRQ	RO - -	0x0

10	MH_RX_FQ2_IRQ	MH interrupt of the RX FIFO Queue 2. Refer to the description of the MH_RX_FQ0_IRQ	RO - -	0x0
9	MH_RX_FQ1_IRQ	MH interrupt of the RX FIFO Queue 1. Refer to the description of the MH_RX_FQ0_IRQ	RO - -	0x0
8	MH_RX_FQ0_IRQ	MH interrupt of the RX FIFO Queue 0. This interrupt is triggered when an invalid RX descriptor is fetched from this RX FIFO Queue, or an RX message is received (if set in RX descriptor) in this RX FIFO Queue, see description of RX_FQ_IRQ[7:0] in MH section.	RO - -	0x0
7	MH_TX_FQ7_IRQ	MH interrupt of the TX FIFO Queue 7. Refer to the description of the MH_TX_FQ0_IRQ	RO - -	0x0
6	MH_TX_FQ6_IRQ	MH interrupt of the TX FIFO Queue 6. Refer to the description of the MH_TX_FQ0_IRQ	RO - -	0x0
5	MH_TX_FQ5_IRQ	MH interrupt of the TX FIFO Queue 5. Refer to the description of the MH_TX_FQ0_IRQ	RO - -	0x0
4	MH_TX_FQ4_IRQ	MH interrupt of the TX FIFO Queue 4. Refer to the description of the MH_TX_FQ0_IRQ	RO - -	0x0
3	MH_TX_FQ3_IRQ	MH interrupt of the TX FIFO Queue 3. Refer to the description of the MH_TX_FQ0_IRQ	RO - -	0x0
2	MH_TX_FQ2_IRQ	MH interrupt of the TX FIFO Queue 2. Refer to the description of the MH_TX_FQ0_IRQ	RO - -	0x0
1	MH_TX_FQ1_IRQ	MH interrupt of the TX FIFO Queue 1. Refer to the description of the MH_TX_FQ0_IRQ	RO - -	0x0
0	MH_TX_FQ0_IRQ	MH interrupt of the TX FIFO Queue 0. This interrupt is triggered when an invalid TX descriptor is fetched from this TX FIFO Queue, a TX message from that FIFO Queue is sent (if set in TX descriptor), or a TX message of that TX FIFO Queue is skipped, see description of TX_FQ_IRQ[7:0] in MH section.	RO - -	0x0

Register: ERR_RAW

Description	: Error raw event status register. This register provides information about the occurrence of functional error relevant events inside the MH and the PRT. A flag is set when the related event is detected, independent of ERR_ENA. The flags remain set until the Host CPU clears them by writing a 1 to the corresponding bit position at register ERR_CLR.
Offset	: 0x4
Absolute Address	: 0xA04
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			TOP_MUX_TO_ERR					PRT_BUS_OFF	PRT_E_PASSIVE	PRT_BUS_ERR	PRT_IFF_RQ	PRT_RX_DO	PRT_TX_DU	PRT_USOS	PRT_ABORTED
			RO - 0x0					RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
		MH_MEM_TO_ERR	MH_WR_RESP_ERR	MH_RD_RESP_ERR	MH_DMA_CH_ERR	MH_DMA_TO_ERR	MH_DP_TO_ERR	MH_DP_DO_ERR	MH_DP_SEQ_ERR	MH_DP_PARITY_ERR	MH_AP_PARITY_ERR	MH_DESC_ERR	MH_REG_CRC_ERR	MH_MEM_SFTY_ERR	MH_RX_FILTER_ERR
		RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
28	TOP_MUX_TO_ERR	Timeout at top-level multiplexer for HOST_AXI detected.	RO - -	0x0
23	PRT_BUS_OFF	PRT entered Bus_Off state.	RO - -	0x0
22	PRT_E_PASSIVE	PRT switched from Error-Active to Error-Passive state.	RO - -	0x0
21	PRT_BUS_ERR	PRT detected error on the CAN Bus.	RO - -	0x0
20	PRT_IFF_RQ	PRT detected invalid Frame Format at TX_MSG.	RO - -	0x0
19	PRT_RX_DO	PRT detected overflow condition at RX_MSG sequence.	RO - -	0x0

18	PRT_TX_DU	PRT detected underrun condition at TX_MSG sequence.	RO - -	0x0
17	PRT_USOS	PRT detected unexpected Start of Sequence during TX_MSG sequence.	RO - -	0x0
16	PRT_ABORTED	PRT detected stop of TX_MSG sequence by TX_MSG_WUSER code ABORT.	RO - -	0x0
13	MH_MEM_TO_ERR	MH detected timeout at local memory interface MEM_AXI, see description of MEM_TO_ERR in MH section.	RO - -	0x0
12	MH_WR_RESP_ERR	MH detected a bus error caused by a write access to S_MEM respective L_MEM, see description of RESP_ERR in MH section.	RO - -	0x0
11	MH_RD_RESP_ERR	MH detected a bus error caused by a read access to S_MEM respective L_MEM, see description of RESP_ERR in MH section.	RO - -	0x0
10	MH_DMA_CH_ERR	MH detected routing error, i.e. data received or sent are not properly routed to or from DMA channel interfaces, see description of DMA_CH_ERR in MH section.	RO - -	0x0
9	MH_DMA_TO_ERR	MH detected timeout at DMA_AXI interface, see description of DMA_TO_ERR in MH section.	RO - -	0x0
8	MH_DP_TO_ERR	MH detected timeout at TX_MSG interface located between MH and PRT, see description of DP_TO_ERR in MH section.	RO - -	0x0
7	MH_DP_DO_ERR	MH detected a data overflow at RX buffer, see description of DP_DO_ERR in MH section.	RO - -	0x0
6	MH_DP_SEQ_ERR	MH detected an incorrect sequence at RX_MSG respective TX_MSG interfaces located between MH and PRT. Associated information provided by MH register ERR_INT_STS, e.g. if RX or TX interface was affected.	RO - -	0x0
5	MH_DP_PARITY_ERR	MH detected parity error at RX message data (received from PRT and written to AXI system bus) respective parity error detected at TX message data (read from AXI system bus and transferred to PRT). Associated information provided by MH register ERR_INT_STS, e.g. if RX message or TX message was affected.	RO - -	0x0
4	MH_AP_PARITY_ERR	MH detected parity error at address pointers, used to manage the MH Queues (RX/TX FIFO Queues and TX Priority Queues). See also description of AP_PARITY_ERR in MH section.	RO - -	0x0

3	MH_DESC_ERR	CRC error detected on RX/TX descriptor or RX/TX descriptor not expected detected. A status flag can define if it is on TX or RX path, see SFTY_INT_STS register.	RO - -	0x0
2	MH_REG_CRC_ERR	MH detected CRC error at the register bank. See also description of REG_CRC_ERR in MH section.	RO - -	0x0
1	MH_MEM_SFTY_ERR	MH detected error in L_MEM. This interrupt is triggered when either the MEM_SFTY_CE or MEM_SFTY_UE input signal is active. The Message Handler provides the information, which signal was active, see flags MH:SFTY_INT_STS.MEM_SFTY_CE and MH:SFTY_INT_STS.MEM_SFTY_UE.	RO - -	0x0
0	MH_RX_FILTER_ERR	MH RX filtering has not finished in time, i.e. current RX filtering has not been completed before next incoming RX message requires RX filtering.	RO - -	0x0

Register: SAFETY_RAW

Description	: Safety raw event status register. This register provides information about the occurrence of safety relevant events inside the MH and the PRT. A flag is set when the related event is detected, independent of SAFETY_ENA. The flags remain set until the Host CPU clears them by writing a 1 to the corresponding bit position at register SAFETY_CLR.
Offset	: 0x8
Absolute Address	: 0xA08
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			TOP_MUX_TO_ERR					PRT_BUS_OFF	PRT_E_PASSIVE	PRT_BUS_ERR	PRT_IFF_RQ	PRT_RX_DO	PRT_TX_DU	PRT_USOS	PRT_ABORTED
			RO - 0x0					RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		MH_MEM_TO_ERR	MH_WR_RESP_ERR	MH_RD_RESP_ERR	MH_DMA_CH_ERR	MH_DMA_TO_ERR	MH_DP_TO_ERR	MH_DP_DO_ERR	MH_DP_SEQ_ERR	MH_DP_PARITY_ERR	MH_AP_PARITY_ERR	MH_DESC_ERR	MH_REG_CRC_ERR	MH_MEM_SFTY_ERR	MH_RX_FILTER_ERR
		RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0	RO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
28	TOP_MUX_TO_ERR	Timeout at top-level multiplexer for HOST_AXI detected.	RO - -	0x0
23	PRT_BUS_OFF	PRT entered Bus_Off state.	RO - -	0x0
22	PRT_E_PASSIVE	PRT switched from Error-Active to Error-Passive state.	RO - -	0x0
21	PRT_BUS_ERR	PRT detected error on the CAN Bus.	RO - -	0x0
20	PRT_IFF_RQ	PRT detected invalid Frame Format at TX_MSG.	RO - -	0x0
19	PRT_RX_DO	PRT detected overflow condition at RX_MSG sequence.	RO - -	0x0
18	PRT_TX_DU	PRT detected underrun condition at TX_MSG sequence.	RO - -	0x0
17	PRT_USOS	PRT detected unexpected Start of Sequence during TX_MSG sequence.	RO - -	0x0
16	PRT_ABORTED	PRT detected stop of TX_MSG sequence by TX_MSG_WUSER code ABORT.	RO - -	0x0
13	MH_MEM_TO_ERR	MH detected timeout at local memory interface MEM_AXI, see description of MEM_TO_ERR in MH section.	RO - -	0x0
12	MH_WR_RESP_ERR	MH detected a bus error caused by a write access to S_MEM respective L_MEM, see description of RESP_ERR in MH section.	RO - -	0x0
11	MH_RD_RESP_ERR	MH detected a bus error caused by a read access to S_MEM respective L_MEM, see description of RESP_ERR in MH section.	RO - -	0x0
10	MH_DMA_CH_ERR	MH detected routing error, i.e. data received or sent are not properly routed to or from DMA channel interfaces, see description of DMA_CH_ERR in MH section.	RO - -	0x0
9	MH_DMA_TO_ERR	MH detected timeout at DMA_AXI interface, see description of DMA_TO_ERR in MH section.	RO - -	0x0
8	MH_DP_TO_ERR	MH detected timeout at TX_MSG interface located between MH and PRT, see description of DP_TO_ERR in MH section.	RO - -	0x0

7	MH_DP_DO_ERR	MH detected a data overflow at RX buffer, see description of DP_DO_ERR in MH section.	RO - -	0x0
6	MH_DP_SEQ_ERR	MH detected an incorrect sequence at RX_MSG respective TX_MSG interfaces located between MH and PRT. Associated information provided by MH register ERR_INT_STS, e.g. if RX or TX interface was affected.	RO - -	0x0
5	MH_DP_PARITY_ERR	MH detected parity error at RX message data (received from PRT and written to AXI system bus) respective parity error detected at TX message data (read from AXI system bus and transferred to PRT). Associated information provided by MH register ERR_INT_STS, e.g. if RX message or TX message was affected.	RO - -	0x0
4	MH_AP_PARITY_ERR	MH detected parity error at address pointers, used to manage the MH Queues (RX/TX FIFO Queues and TX Priority Queues). See also description of AP_PARITY_ERR in MH section.	RO - -	0x0
3	MH_DESC_ERR	CRC error detected on RX/TX descriptor or RX/TX descriptor not expected detected. A status flag can define if it is on TX or RX path, see SFTY_INT_STS register.	RO - -	0x0
2	MH_REG_CRC_ERR	MH detected CRC error at the register bank. See also description of REG_CRC_ERR in MH section.	RO - -	0x0
1	MH_MEM_SFTY_ERR	MH detected error in L_MEM. This interrupt is triggered when either the MEM_SFTY_CE or MEM_SFTY_UE input signal is active. The Message Handler provides the information, which signal was active, see flags MH:SFTY_INT_STS.MEM_SFTY_CE and MH:SFTY_INT_STS.MEM_SFTY_UE.	RO - -	0x0
0	MH_RX_FILTER_ERR	MH RX filtering has not finished in time, i.e. current RX filtering has not been completed before next incoming RX message requires RX filtering.	RO - -	0x0

Block: xcand_top_irc_inst_0__irc_axi__a00__Control

Description : This address block provides the registers for controlling the IRC.

Register: FUNC_CLR

Description	: Functional raw event clear register. Writing a 1 to a certain bit position clears the corresponding bit of register FUNC_RAW. Writing a ’0’ has no effect.
Offset	: 0x0
Absolute Address	: 0xA10
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				PRT_RX_EVT	PRT_TX_EVT	PRT_BUS_ON	PRT_E_ACTIVE		MH_STATS_IRQ	MH_RX_ABORT_IRQ	MH_TX_ABORT_IRQ	MH_TX_FILTER_IRQ	MH_RX_FILTER_IRQ	MH_STOP_IRQ	MH_TX_PQ_IRQ
				WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0		WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MH_RX_FQ7_IRQ	MH_RX_FQ6_IRQ	MH_RX_FQ5_IRQ	MH_RX_FQ4_IRQ	MH_RX_FQ3_IRQ	MH_RX_FQ2_IRQ	MH_RX_FQ1_IRQ	MH_RX_FQ0_IRQ	MH_TX_FQ7_IRQ	MH_TX_FQ6_IRQ	MH_TX_FQ5_IRQ	MH_TX_FQ4_IRQ	MH_TX_FQ3_IRQ	MH_TX_FQ2_IRQ	MH_TX_FQ1_IRQ	MH_TX_FQ0_IRQ
WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
27	PRT_RX_EVT	Clear bit FUNC_RAW.PRT_RX_EVT by writing 1	WO - -	0x0
26	PRT_TX_EVT	Clear bit FUNC_RAW.PRT_TX_EVT by writing 1	WO - -	0x0
25	PRT_BUS_ON	Clear bit FUNC_RAW.PRT_BUS_ON by writing 1	WO - -	0x0

24	PRT_E_ACTIVE	Clear bit FUNC_RAW.PRT_E_ACTIVE by writing 1	WO - -	0x0
22	MH_STATS_IRQ	Clear bit FUNC_RAW.MH_STATS_IRQ by writing 1	WO - -	0x0
21	MH_RX_ABORT_IRQ	Clear bit FUNC_RAW.MH_RX_ABORT_IRQ by writing 1	WO - -	0x0
20	MH_TX_ABORT_IRQ	Clear bit FUNC_RAW.MH_TX_ABORT_IRQ by writing 1	WO - -	0x0
19	MH_TX_FILTER_IRQ	Clear bit FUNC_RAW.MH_TX_FILTER_IRQ by writing 1	WO - -	0x0
18	MH_RX_FILTER_IRQ	Clear bit FUNC_RAW.MH_RX_FILTER_IRQ by writing 1	WO - -	0x0
17	MH_STOP_IRQ	Clear bit FUNC_RAW.MH_STOP_IRQ by writing 1	WO - -	0x0
16	MH_TX_PQ_IRQ	Clear bit FUNC_RAW.MH_TX_PQ_IRQ by writing 1	WO - -	0x0
15	MH_RX_FQ7_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ7_IRQ by writing 1	WO - -	0x0
14	MH_RX_FQ6_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ6_IRQ by writing 1	WO - -	0x0
13	MH_RX_FQ5_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ5_IRQ by writing 1	WO - -	0x0
12	MH_RX_FQ4_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ4_IRQ by writing 1	WO - -	0x0
11	MH_RX_FQ3_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ3_IRQ by writing 1	WO - -	0x0
10	MH_RX_FQ2_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ2_IRQ by writing 1	WO - -	0x0
9	MH_RX_FQ1_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ1_IRQ by writing 1	WO - -	0x0
8	MH_RX_FQ0_IRQ	Clear bit of FUNC_RAW.MH_RX_FQ0_IRQ by writing 1	WO - -	0x0
7	MH_TX_FQ7_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ7_IRQ by writing 1	WO - -	0x0

6	MH_TX_FQ6_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ6_IRQ by writing 1	WO - -	0x0
5	MH_TX_FQ5_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ5_IRQ by writing 1	WO - -	0x0
4	MH_TX_FQ4_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ4_IRQ by writing 1	WO - -	0x0
3	MH_TX_FQ3_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ3_IRQ by writing 1	WO - -	0x0
2	MH_TX_FQ2_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ2_IRQ by writing 1	WO - -	0x0
1	MH_TX_FQ1_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ1_IRQ by writing 1	WO - -	0x0
0	MH_TX_FQ0_IRQ	Clear bit of FUNC_RAW.MH_TX_FQ0_IRQ by writing 1	WO - -	0x0

Register: ERR_CLR

Description	: Error raw event clear register. Writing a 1 to a certain bit position clears the corresponding bit of register ERR_RAW. Writing a '0' has no effect.
Offset	: 0x4
Absolute Address	: 0xA14
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			TOP_MUX_TO_ERR					PRT_BUS_OFF	PRT_E_PASSIVE	PRT_BUS_ERR	PRT_IFF_RQ	PRT_RX_DO	PRT_TX_DU	PRT_USOS	PRT_ABORTED
			WO - 0x0					WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		MH_MEM_TO_ERR	MH_WR_RESP_ERR	MH_RD_RESP_ERR	MH_DMA_CH_ERR	MH_DMA_TO_ERR	MH_DP_TO_ERR	MH_DP_DO_ERR	MH_DP_SEQ_ERR	MH_DP_PARITY_ERR	MH_AP_PARITY_ERR	MH_DESC_ERR	MH_REG_CRC_ERR	MH_MEM_SFTY_ERR	MH_RX_FILTER_ERR
		WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
28	TOP_MUX_TO_ERR	Clear bit ERR_RAW.TOP_MUX_TO_ERR by writing 1	WO - -	0x0
23	PRT_BUS_OFF	Clear bit ERR_RAW.PRT_BUS_OFF by writing 1	WO - -	0x0
22	PRT_E_PASSIVE	Clear bit ERR_RAW.PRT_E_PASSIVE by writing 1	WO - -	0x0
21	PRT_BUS_ERR	Clear bit ERR_RAW.PRT_BUS_ERR by writing 1	WO - -	0x0
20	PRT_IFF_RQ	Clear bit ERR_RAW.PRT_IFF_RQ by writing 1	WO - -	0x0
19	PRT_RX_DO	Clear bit ERR_RAW.PRT_RX_DO by writing 1	WO - -	0x0
18	PRT_TX_DU	Clear bit ERR_RAW.PRT_TX_DU by writing 1	WO - -	0x0
17	PRT_USOS	Clear bit ERR_RAW.PRT_USOS by writing 1	WO - -	0x0
16	PRT_ABORTED	Clear bit ERR_RAW.PRT_ABORTED by writing 1	WO - -	0x0
13	MH_MEM_TO_ERR	Clear bit ERR_RAW.MH_MEM_TO_ERR by writing 1	WO - -	0x0
12	MH_WR_RESP_ERR	Clear bit ERR_RAW.MH_WR_RESP_ERR by writing 1	WO - -	0x0
11	MH_RD_RESP_ERR	Clear bit ERR_RAW.MH_RD_RESP_ERR by writing 1	WO - -	0x0
10	MH_DMA_CH_ERR	Clear bit ERR_RAW.MH_DMA_CH_ERR by writing 1	WO - -	0x0
9	MH_DMA_TO_ERR	Clear bit ERR_RAW.MH_DMA_TO_ERR by writing 1	WO - -	0x0
8	MH_DP_TO_ERR	Clear bit ERR_RAW.MH_DP_TO_ERR by writing 1	WO - -	0x0

7	MH_DP_DO_ERR	Clear bit ERR_RAW.MH_DP_DO_ERR by writing 1	WO - -	0x0
6	MH_DP_SEQ_ERR	Clear bit ERR_RAW.MH_DP_SEQ_ERR by writing 1	WO - -	0x0
5	MH_DP_PARITY_ERR	Clear bit ERR_RAW.MH_DP_PARITY_ERR by writing 1	WO - -	0x0
4	MH_AP_PARITY_ERR	Clear bit ERR_RAW.MH_AP_PARITY_ERR by writing 1	WO - -	0x0
3	MH_DESC_ERR	Clear bit ERR_RAW.MH_DESC_ERR by writing 1	WO - -	0x0
2	MH_REG_CRC_ERR	Clear bit ERR_RAW.MH_REG_CRC_ERR by writing 1	WO - -	0x0
1	MH_MEM_SFTY_ERR	Clear bit ERR_RAW.MH_MEM_SFTY_ERR by writing 1	WO - -	0x0
0	MH_RX_FILTER_ERR	Clear bit ERR_RAW.MH_RX_FILTER_ERR by writing 1	WO - -	0x0

Register: SAFETY_CLR

Description	: Safety raw event clear register. Writing a 1 to a certain bit position clears the corresponding bit of register SAFETY_RAW. Writing a ’0’ has no effect.
Offset	: 0x8
Absolute Address	: 0xA18
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			TOP_MUX_TO_ERR					PRT_BUS_OFF	PRT_E_PASSIVE	PRT_BUS_ERR	PRT_IFF_RQ	PRT_RX_DO	PRT_TX_DU	PRT_USOS	PRT_ABORTED
			WO - 0x0					WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		MH_MEM_TO_ERR	MH_WR_RESP_ERR	MH_RD_RESP_ERR	MH_DMA_CH_ERR	MH_DMA_TO_ERR	MH_DP_TO_ERR	MH_DP_DO_ERR	MH_DP_SEQ_ERR	MH_DP_PARITY_ERR	MH_AP_PARITY_ERR	MH_DESC_ERR	MH_REG_CRC_ERR	MH_MEM_SFTY_ERR	MH_RX_FILTER_ERR

	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0	WO - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
28	TOP_MUX_TO_ERR	Clear bit SAFETY_RAW.TOP_MUX_TO_ERR by writing 1	WO - -	0x0
23	PRT_BUS_OFF	Clear bit SAFETY_RAW.PRT_BUS_OFF by writing 1	WO - -	0x0
22	PRT_E_PASSIVE	Clear bit SAFETY_RAW.PRT_E_PASSIVE by writing 1	WO - -	0x0
21	PRT_BUS_ERR	Clear bit SAFETY_RAW.PRT_BUS_ERR by writing 1	WO - -	0x0
20	PRT_IFF_RQ	Clear bit SAFETY_RAW.PRT_IFF_RQ by writing 1	WO - -	0x0
19	PRT_RX_DO	Clear bit SAFETY_RAW.PRT_RX_DO by writing 1	WO - -	0x0
18	PRT_TX_DU	Clear bit SAFETY_RAW.PRT_TX_DU by writing 1	WO - -	0x0
17	PRT_USOS	Clear bit SAFETY_RAW.PRT_USOS by writing 1	WO - -	0x0
16	PRT_ABORTED	Clear bit SAFETY_RAW.PRT_ABORTED by writing 1	WO - -	0x0
13	MH_MEM_TO_ERR	Clear bit SAFETY_RAW.MH_MEM_TO_ERR by writing 1	WO - -	0x0
12	MH_WR_RESP_ERR	Clear bit SAFETY_RAW.MH_WR_RESP_ERR by writing 1	WO - -	0x0
11	MH_RD_RESP_ERR	Clear bit SAFETY_RAW.MH_RD_RESP_ERR by writing 1	WO - -	0x0
10	MH_DMA_CH_ERR	Clear bit SAFETY_RAW.MH_DMA_CH_ERR by writing 1	WO - -	0x0
9	MH_DMA_TO_ERR	Clear bit SAFETY_RAW.MH_DMA_TO_ERR by writing 1	WO - -	0x0

8	MH_DP_TO_ERR	Clear bit SAFETY_RAW.MH_DP_TO_ERR by writing 1	WO - -	0x0
7	MH_DP_DO_ERR	Clear bit SAFETY_RAW.MH_DP_DO_ERR by writing 1	WO - -	0x0
6	MH_DP_SEQ_ERR	Clear bit SAFETY_RAW.MH_DP_SEQ_ERR by writing 1	WO - -	0x0
5	MH_DP_PARITY_ERR	Clear bit SAFETY_RAW.MH_DP_PARITY_ERR by writing 1	WO - -	0x0
4	MH_AP_PARITY_ERR	Clear bit SAFETY_RAW.MH_AP_PARITY_ERR by writing 1	WO - -	0x0
3	MH_DESC_ERR	Clear bit SAFETY_RAW.MH_DESC_ERR by writing 1	WO - -	0x0
2	MH_REG_CRC_ERR	Clear bit SAFETY_RAW.MH_REG_CRC_ERR by writing 1	WO - -	0x0
1	MH_MEM_SFTY_ERR	Clear bit SAFETY_RAW.MH_MEM_SFTY_ERR by writing 1	WO - -	0x0
0	MH_RX_FILTER_ERR	Clear bit SAFETY_RAW.MH_RX_FILTER_ERR by writing 1	WO - -	0x0

Register: FUNC_ENA

Description	: Functional raw event enable register. Any bit in the FUNC_ENA register enables the corresponding bit in the FUNC_RAW to trigger the interrupt line FUNC_INT. The interrupt line gets active high, when at least one RAW/ENA pair is 1, e.g. FUNC_RAW.MH_TX_FQ_IRQ = FUNC_ENA.MH_TX_FQ_IRQ = 1
Offset	: 0x10
Absolute Address	: 0xA20
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
				PRT_RX_EVT	PRT_TX_EVT	PRT_BUS_ON	PRT_E_ACTIVE		MH_STATS_IRQ	MH_RX_ABORT_IRQ	MH_TX_ABORT_IRQ	MH_TX_FILTER_IRQ	MH_RX_FILTER_IRQ	MH_STOP_IRQ	MH_TX_PQ_IRQ
				RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0		RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

MH_RX_FQ7_IRQ	MH_RX_FQ6_IRQ	MH_RX_FQ5_IRQ	MH_RX_FQ4_IRQ	MH_RX_FQ3_IRQ	MH_RX_FQ2_IRQ	MH_RX_FQ1_IRQ	MH_RX_FQ0_IRQ	MH_TX_FQ7_IRQ	MH_TX_FQ6_IRQ	MH_TX_FQ5_IRQ	MH_TX_FQ4_IRQ	MH_TX_FQ3_IRQ	MH_TX_FQ2_IRQ	MH_TX_FQ1_IRQ	MH_TX_FQ0_IRQ
RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
27	PRT_RX_EVT	Enable FUNC_RAW.PRT_RX_EVT to activate FUNC_INT	RW - -	0x0
26	PRT_TX_EVT	Enable FUNC_RAW.PRT_TX_EVT to activate FUNC_INT	RW - -	0x0
25	PRT_BUS_ON	Enable FUNC_RAW.PRT_BUS_ON to activate FUNC_INT	RW - -	0x0
24	PRT_E_ACTIVE	Enable FUNC_RAW.PRT_E_ACTIVE to activate FUNC_INT	RW - -	0x0
22	MH_STATS_IRQ	Enable FUNC_RAW.MH_STATS_IRQ to activate FUNC_INT	RW - -	0x0
21	MH_RX_ABORT_IRQ	Enable FUNC_RAW.MH_RX_ABORT_IRQ to activate FUNC_INT	RW - -	0x0
20	MH_TX_ABORT_IRQ	Enable FUNC_RAW.MH_TX_ABORT_IRQ to activate FUNC_INT	RW - -	0x0
19	MH_TX_FILTER_IRQ	Enable FUNC_RAW.MH_TX_FILTER_IRQ to activate FUNC_INT	RW - -	0x0
18	MH_RX_FILTER_IRQ	Enable FUNC_RAW.MH_RX_FILTER_IRQ to activate FUNC_INT	RW - -	0x0
17	MH_STOP_IRQ	Enable FUNC_RAW.MH_STOP_IRQ to activate FUNC_INT	RW - -	0x0
16	MH_TX_PQ_IRQ	Enable FUNC_RAW.MH_TX_PQ_IRQ to activate FUNC_INT	RW - -	0x0
15	MH_RX_FQ7_IRQ	Enable FUNC_RAW.MH_RX_FQ7_IRQ to activate FUNC_INT	RW - -	0x0
14	MH_RX_FQ6_IRQ	Enable FUNC_RAW.MH_RX_FQ6_IRQ to activate FUNC_INT	RW - -	0x0

13	MH_RX_ FQ5_IRQ	Enable FUNC_RAW.MH_RX_FQ5_IRQ to activate FUNC_INT	RW - -	0x0
12	MH_RX_ FQ4_IRQ	Enable FUNC_RAW.MH_RX_FQ4_IRQ to activate FUNC_INT	RW - -	0x0
11	MH_RX_ FQ3_IRQ	Enable FUNC_RAW.MH_RX_FQ3_IRQ to activate FUNC_INT	RW - -	0x0
10	MH_RX_ FQ2_IRQ	Enable FUNC_RAW.MH_RX_FQ2_IRQ to activate FUNC_INT	RW - -	0x0
9	MH_RX_ FQ1_IRQ	Enable FUNC_RAW.MH_RX_FQ1_IRQ to activate FUNC_INT	RW - -	0x0
8	MH_RX_ FQ0_IRQ	Enable FUNC_RAW.MH_RX_FQ0_IRQ to activate FUNC_INT	RW - -	0x0
7	MH_TX_ FQ7_IRQ	Enable FUNC_RAW.MH_TX_FQ7_IRQ to activate FUNC_INT	RW - -	0x0
6	MH_TX_ FQ6_IRQ	Enable FUNC_RAW.MH_TX_FQ6_IRQ to activate FUNC_INT	RW - -	0x0
5	MH_TX_ FQ5_IRQ	Enable FUNC_RAW.MH_TX_FQ5_IRQ to activate FUNC_INT	RW - -	0x0
4	MH_TX_ FQ4_IRQ	Enable FUNC_RAW.MH_TX_FQ4_IRQ to activate FUNC_INT	RW - -	0x0
3	MH_TX_ FQ3_IRQ	Enable FUNC_RAW.MH_TX_FQ3_IRQ to activate FUNC_INT	RW - -	0x0
2	MH_TX_ FQ2_IRQ	Enable FUNC_RAW.MH_TX_FQ2_IRQ to activate FUNC_INT	RW - -	0x0
1	MH_TX_ FQ1_IRQ	Enable FUNC_RAW.MH_TX_FQ1_IRQ to activate FUNC_INT	RW - -	0x0
0	MH_TX_ FQ0_IRQ	Enable FUNC_RAW.MH_TX_FQ0_IRQ to activate FUNC_INT	RW - -	0x0

Register: ERR_ENA

Description	: Error raw event enable register. Any bit in the ERR_ENA register enables the corresponding bit in the ERR_RAW to trigger the interrupt line ERR_INT. The interrupt line gets active high, when at least one RAW/ENA pair is 1, e.g. ERR_RAW.MH_TX_FQ_IRQ = ERR_ENA.MH_TX_FQ_IRQ = 1
Offset	: 0x14
Absolute Address	: 0xA24
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			TOP_MUX_TO_ERR					PRT_BUS_OFF	PRT_E_PASSIVE	PRT_BUS_ERR	PRT_IFF_RQ	PRT_RX_DO	PRT_TX_DU	PRT_USOS	PRT_ABORTED
			RW - 0x0					RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
		MH_MEM_TO_ERR	MH_WR_RESP_ERR	MH_RD_RESP_ERR	MH_DMA_CH_ERR	MH_DMA_TO_ERR	MH_DP_TO_ERR	MH_DP_DO_ERR	MH_DP_SEQ_ERR	MH_DP_PARITY_ERR	MH_AP_PARITY_ERR	MH_DESC_ERR	MH_REG_CRC_ERR	MH_MEM_SFTY_ERR	MH_RX_FILTER_ERR
		RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
28	TOP_MUX_TO_ERR	Enable ERR_RAW.TOP_MUX_TO_ERR to activate ERR_INT	RW - -	0x0
23	PRT_BUS_OFF	Enable ERR_RAW.PRT_BUS_OFF to activate ERR_INT	RW - -	0x0
22	PRT_E_PASSIVE	Enable ERR_RAW.PRT_E_PASSIVE to activate ERR_INT	RW - -	0x0
21	PRT_BUS_ERR	Enable ERR_RAW.PRT_BUS_ERR to activate ERR_INT	RW - -	0x0
20	PRT_IFF_RQ	Enable ERR_RAW.PRT_IFF_RQ to activate ERR_INT	RW - -	0x0
19	PRT_RX_DO	Enable ERR_RAW.PRT_RX_DO to activate ERR_INT	RW - -	0x0

18	PRT_TX_DU	Enable ERR_RAW.PRT_TX_DU to activate ERR_INT	RW - -	0x0
17	PRT_USOS	Enable ERR_RAW.PRT_USOS to activate ERR_INT	RW - -	0x0
16	PRT_ABORTED	Enable ERR_RAW.PRT_ABORTED to activate ERR_INT	RW - -	0x0
13	MH_MEM_TO_ERR	Enable ERR_RAW.MH_MEM_TO_ERR to activate ERR_INT	RW - -	0x0
12	MH_WR_RESP_ERR	Enable ERR_RAW.MH_WR_RESP_ERR to activate ERR_INT	RW - -	0x0
11	MH_RD_RESP_ERR	Enable ERR_RAW.MH_RD_RESP_ERR to activate ERR_INT	RW - -	0x0
10	MH_DMA_CH_ERR	Enable ERR_RAW.MH_DMA_CH_ERR to activate ERR_INT	RW - -	0x0
9	MH_DMA_TO_ERR	Enable ERR_RAW.MH_DMA_TO_ERR to activate ERR_INT	RW - -	0x0
8	MH_DP_TO_ERR	Enable ERR_RAW.MH_DP_TO_ERR to activate ERR_INT	RW - -	0x0
7	MH_DP_DO_ERR	Enable ERR_RAW.MH_DP_DO_ERR to activate ERR_INT	RW - -	0x0
6	MH_DP_SEQ_ERR	Enable ERR_RAW.MH_DP_SEQ_ERR to activate ERR_INT	RW - -	0x0
5	MH_DP_PARITY_ERR	Enable ERR_RAW.MH_DP_PARITY_ERR to activate ERR_INT	RW - -	0x0
4	MH_AP_PARITY_ERR	Enable ERR_RAW.MH_AP_PARITY_ERR to activate ERR_INT	RW - -	0x0
3	MH_DESC_ERR	Enable ERR_RAW.MH_DESC_ERR to activate ERR_INT	RW - -	0x0
2	MH_REG_CRC_ERR	Enable ERR_RAW.MH_REG_CRC_ERR to activate ERR_INT	RW - -	0x0
1	MH_MEM_SFTY_ERR	Enable ERR_RAW.MH_MEM_SFTY_ERR to activate ERR_INT	RW - -	0x0
0	MH_RX_FILTER_ERR	Enable ERR_RAW.MH_RX_FILTER_ERR to activate ERR_INT	RW - -	0x0

Register: SAFETY_ENA

Description	: Safety raw event enable register. Any bit in the SAFETY_ENA register enables the corresponding bit in the SAFETY_RAW to trigger the interrupt line SAFETY_INT. The interrupt line gets active high, when at least one RAW/ENA pair is 1, e.g. SAFETY_RAW.MH_TX_FQ_IRQ = SAFETY_ENA.MH_TX_FQ_IRQ = 1
Offset	: 0x18
Absolute Address	: 0xA28
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			TOP_MUX_TO_ERR					PRT_BUS_OFF	PRT_E_PASSIVE	PRT_BUS_ERR	PRT_IFF_RQ	PRT_RX_DO	PRT_TX_DU	PRT_USOS	PRT_ABORTED
			RW - 0x0					RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		MH_MEM_TO_ERR	MH_WR_RESP_ERR	MH_RD_RESP_ERR	MH_DMA_CH_ERR	MH_DMA_TO_ERR	MH_DP_TO_ERR	MH_DP_DO_ERR	MH_DP_SEQ_ERR	MH_DP_PARITY_ERR	MH_AP_PARITY_ERR	MH_DES_C_ERR	MH_REG_CRC_ERR	MH_MEM_SFTY_ERR	MH_RX_FILTER_ERR
		RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0	RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
28	TOP_MUX_TO_ERR	Enable SAFETY_RAW.TOP_MUX_TO_ERR to activate SAFETY_INT	RW - -	0x0
23	PRT_BUS_OFF	Enable SAFETY_RAW.PRT_BUS_OFF to activate SAFETY_INT	RW - -	0x0
22	PRT_E_PASSIVE	Enable SAFETY_RAW.PRT_E_PASSIVE to activate SAFETY_INT	RW - -	0x0
21	PRT_BUS_ERR	Enable SAFETY_RAW.PRT_BUS_ERR to activate SAFETY_INT	RW - -	0x0

20	PRT_IFF_RQ	Enable SAFETY_RAW.PRT_IFF_RQ to activate SAFETY_INT	RW - -	0x0
19	PRT_RX_DO	Enable SAFETY_RAW.PRT_RX_DO to activate SAFETY_INT	RW - -	0x0
18	PRT_TX_DU	Enable SAFETY_RAW.PRT_TX_DU to activate SAFETY_INT	RW - -	0x0
17	PRT_USOS	Enable SAFETY_RAW.PRT_USOS to activate SAFETY_INT	RW - -	0x0
16	PRT_ABORTED	Enable SAFETY_RAW.PRT_ABORTED to activate SAFETY_INT	RW - -	0x0
13	MH_MEM_TO_ERR	Enable SAFETY_RAW.MH_MEM_TO_ERR to activate SAFETY_INT	RW - -	0x0
12	MH_WR_RESP_ERR	Enable SAFETY_RAW.MH_WR_RESP_ERR to activate SAFETY_INT	RW - -	0x0
11	MH_RD_RESP_ERR	Enable SAFETY_RAW.MH_RD_RESP_ERR to activate SAFETY_INT	RW - -	0x0
10	MH_DMA_CH_ERR	Enable SAFETY_RAW.MH_DMA_CH_ERR to activate SAFETY_INT	RW - -	0x0
9	MH_DMA_TO_ERR	Enable SAFETY_RAW.MH_DMA_TO_ERR to activate SAFETY_INT	RW - -	0x0
8	MH_DP_TO_ERR	Enable SAFETY_RAW.MH_DP_TO_ERR to activate SAFETY_INT	RW - -	0x0
7	MH_DP_DO_ERR	Enable SAFETY_RAW.MH_DP_DO_ERR to activate SAFETY_INT	RW - -	0x0
6	MH_DP_SEQ_ERR	Enable SAFETY_RAW.MH_DP_SEQ_ERR to activate SAFETY_INT	RW - -	0x0
5	MH_DP_PARITY_ERR	Enable SAFETY_RAW.MH_DP_PARITY_ERR to activate SAFETY_INT	RW - -	0x0
4	MH_AP_PARITY_ERR	Enable SAFETY_RAW.MH_AP_PARITY_ERR to activate SAFETY_INT	RW - -	0x0
3	MH_DESC_ERR	Enable SAFETY_RAW.MH_DESC_ERR to activate SAFETY_INT	RW - -	0x0
2	MH_REG_CRC_ERR	Enable SAFETY_RAW.MH_REG_CRC_ERR to activate SAFETY_INT	RW - -	0x0

1	MH_MEM_SFTY_ERR	Enable SAFETY_RAW.MH_MEM_SFTY_ERR to activate SAFETY_INT	RW - -	0x0
0	MH_RX_FILTER_ERR	Enable SAFETY_RAW.MH_RX_FILTER_ERR to activate SAFETY_INT	RW - -	0x0

Block: xcand_top_irc_inst_0__irc_axi__a00__Configuration

Description	: Hardware configuration of the IRC

Register: CAPTURING_MODE

Description	: IRC configuration register. This register shows the hardware configuration of the IRC concerning the capturing mode of the event inputs. The IP internal events signals coming from the MH and the PRT require an 'edge sensitive' capturing. That is why the value of this register is 0x7 and cannot be changed.
Offset	: 0x0
Absolute Address	: 0xA30
Addressing Mode	: 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
													SAFETY	ERR	FUNC
													RO - 0x1	RO - 0x1	RO - 0x1

Bit	Name	Description	SW Access HW Access Protection	Reset
2	SAFETY	Capturing mode of SAFETY RAW register. 0 = Level sensitive 1 = Edge sensitive	RO - -	0x1
1	ERR	Capturing mode of ERR RAW register. 0 = Level sensitive 1 = Edge sensitive	RO - -	0x1
0	FUNC	Capturing mode of FUNC_RAW register. 0 = Level sensitive 1 = Edge sensitive	RO - -	0x1

Block: xcand_top_irc_inst_0__irc_axi__a00__Auxiliary

Description : Auxiliary registers

Register: HDP

Description : Hardware Debug Port control register
Offset : 0x0
Absolute Address : 0xA40
Addressing Mode : 32-bits

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
															HDP_SEL
															RW - 0x0

Bit	Name	Description	SW Access HW Access Protection	Reset
0	HDP_SEL	Select the driver of the Hardware Debug Port. See also chapter HDP. 0 = Message Handler 1 = Protocol Controller	RW - -	0x0

Access policy

Access policy	Description	Effect of a Write on Current Field Value	Effect of a Read on Current Field Value
RO	Read Only	No effect.	No effect.
RW	Read Write	Changed to written value.	No effect.
RC	Read Clears All	No effect.	Sets all bits to 0.
RS	Read Sets All	No effect.	Sets all bits to 1.
WRC	Write, Read Clears All	Changed to written value.	Sets all bits to 0.
WRS	Write, Read Sets All	Changed to written value.	Sets all bits to 1.
WC	Write Clears All	Sets all bits to 0.	No effect.
WS	Write Sets All	Sets all bits to 1.	No effect.
WSRC	Write Sets All, Read Clears All	Sets all bits to 1.	Sets all bits to 0.
WCRS	Write Clears All, Read Sets All	Sets all bits to 0.	Sets all bits to 1.
W1C	Write 1 to Clear	If the bit in the written value is a 1, the corresponding bit in the field is set to 0. Otherwise, the field bit is not affected.	No effect.
W1S	Write 1 to Set	If the bit in the written value is a 1, the corresponding bit in the field is set to 1. Otherwise, the field bit is not affected.	No effect.
W1T	Write 1 to Toggle	If the bit in the written value is a 1, the corresponding bit in the field is inverted. Otherwise, the field bit is not affected.	No effect.
W0C	Write 0 to Clear	If the bit in the written value is a 0, the corresponding bit in the field is set to 0. Otherwise, the field bit is not affected.	No effect.
W0S	Write 0 to Set	If the bit in the written value is a 0, the corresponding bit in the field is set to 1. Otherwise, the field bit is not affected.	No effect.
W0T	Write 0 to Toggle	If the bit in the written value is a 0, the corresponding bit in the field is inverted. Otherwise, the field bit is not affected.	No effect.
W1SRC	Write 1 to Set, Read Clears All	If the bit in the written value is a 1, the corresponding bit in the field is set to 1. Otherwise, the field bit is not affected.	Sets all bits to 0.
W1CRS	Write 1 to Clear, Read Sets All	If the bit in the written value is a 1, the corresponding bit in the field is set to 0. Otherwise, the field bit is not affected.	Sets all bits to 1.
W0SRC	Write 0 to Set, Read Clears All	If the bit in the written value is a 0, the corresponding bit in the field is set to 1. Otherwise, the field bit is not affected.	Sets all bits to 0.

W0C RS	Write 0 to Clear, Read Sets All	If the bit in the written value is a 0, the corresponding bit in the field is set to 0. Otherwise, the field bit is not affected.	Sets all bits to 1.
W0	Write Only	Changed to written value	No effect.
W0C	Write Only Clears All	Sets all bits to 0.	No effect.
W0S	Write Only Sets All	Sets all bits to 1.	No effect.
W1	Write Once	Changed to written value if this is the first write operation after a hard reset. Otherwise, has no effect.	No effect.
W01	Write Only, Once	Changed to written value if this is the first write operation after a hard reset. Otherwise, has no effect.	No effect.